

Short Course on Wireless Power Transfer Technologies



香港城市大學
City University of Hong Kong

Part IV: Transformer Design

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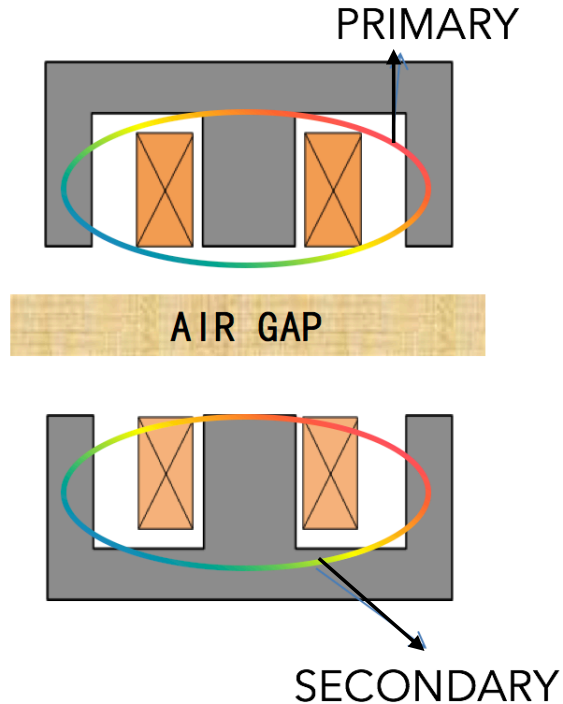
<http://www.ee.cityu.edu.hk/~chitse>



TRANSFORMER DESIGN

PART IV

Contactless Transformers

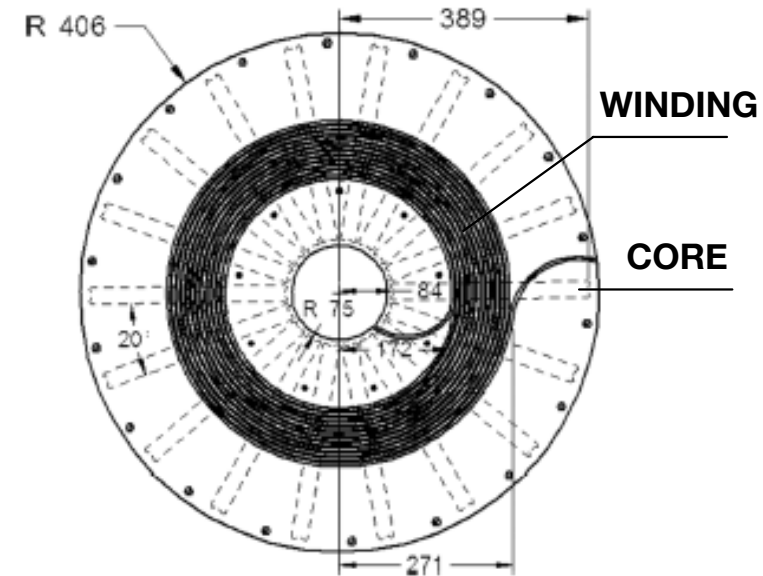


Transformers with low coupling coefficients

- Large leakage inductance
- Small magnetizing inductance

Design Issues

Raise coupling coefficient
Reduce effect of misalignment
Increase Q (less loss)
Reduce volume and size
EMI issues



Circular Pad Transformer

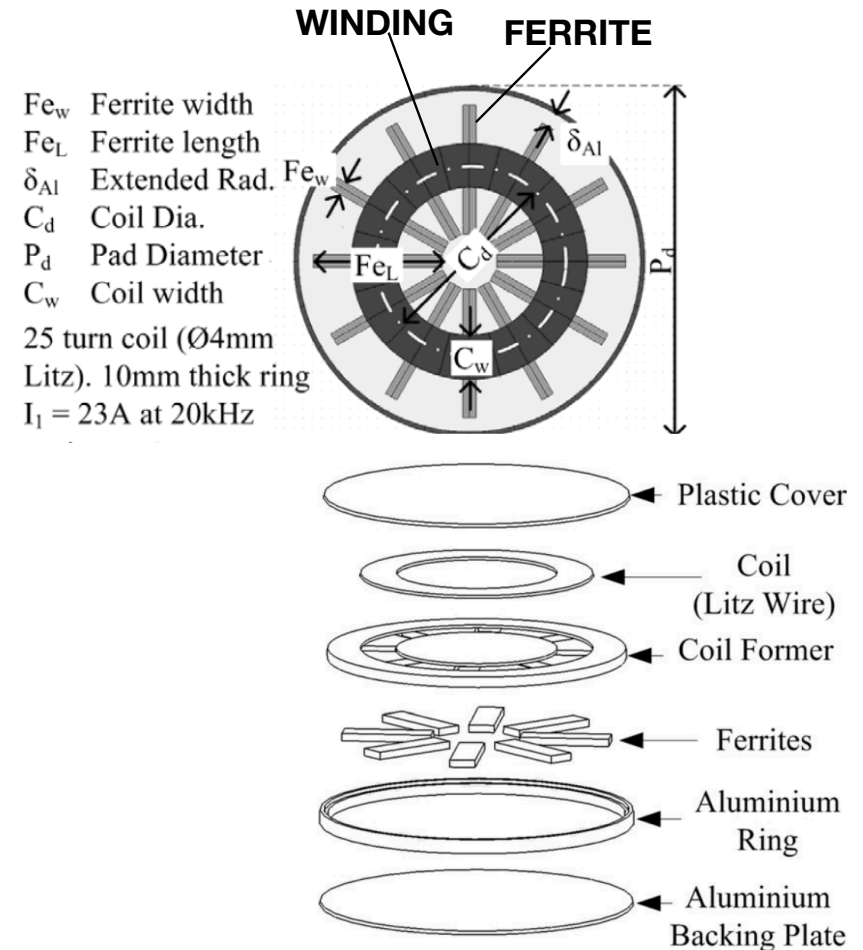
Power Pad (Auckland)

Transformers with stripped cores

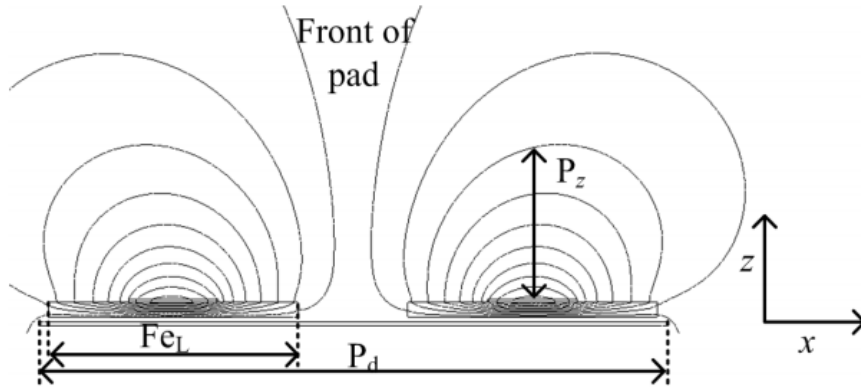
- Coupling coefficient is not seriously affected
- Volume is dramatically reduced

Design Issues

- Winding distributions affect the power transfer capacity measured by P_{su} which is the uncompensated power ($P_{su} = V_{oc}I_{sc}$). So, $P_{out} = P_{su}Q$.
- When [coil diameter]/[pad diameter] ≈ 0.57 , the coupling coefficient and power transfer can be optimal.
- Presence of outer Al ring affects the power transfer capability
- Greater the distance of outer ring to ferrite, better the coupling and power transfer capability

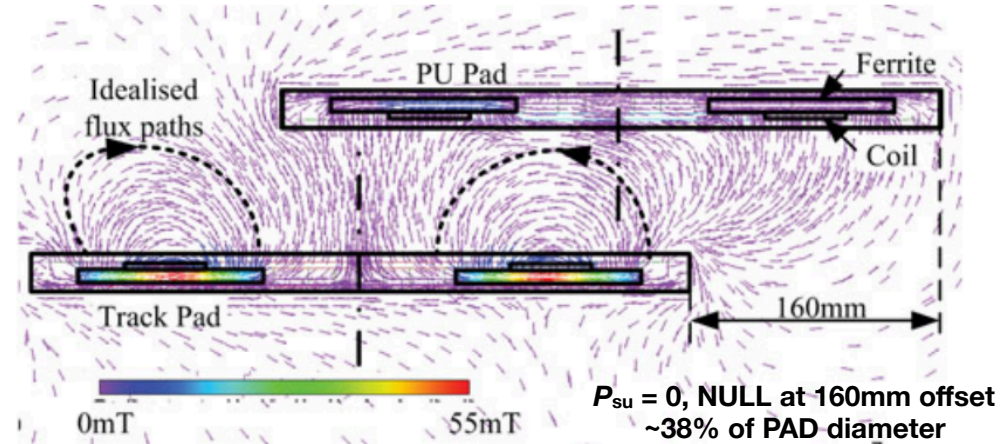


Empirical Data (Univ. of Auckland)

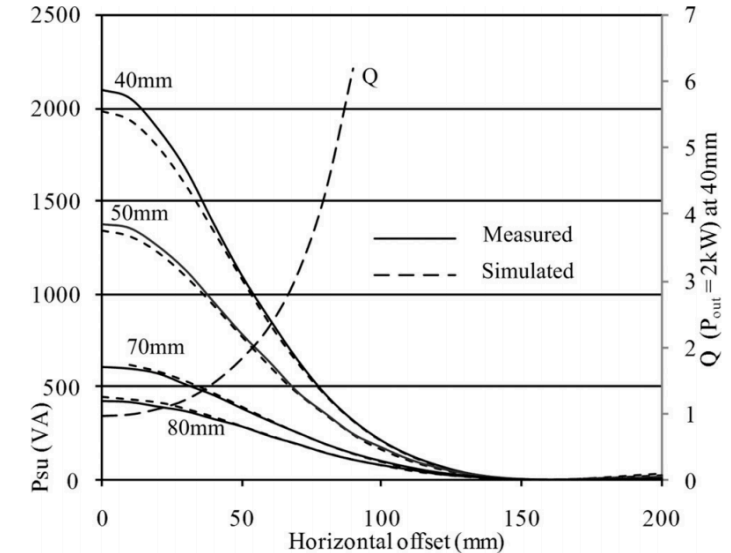


Flux density well below saturation which is around 200 mT for ferrite.

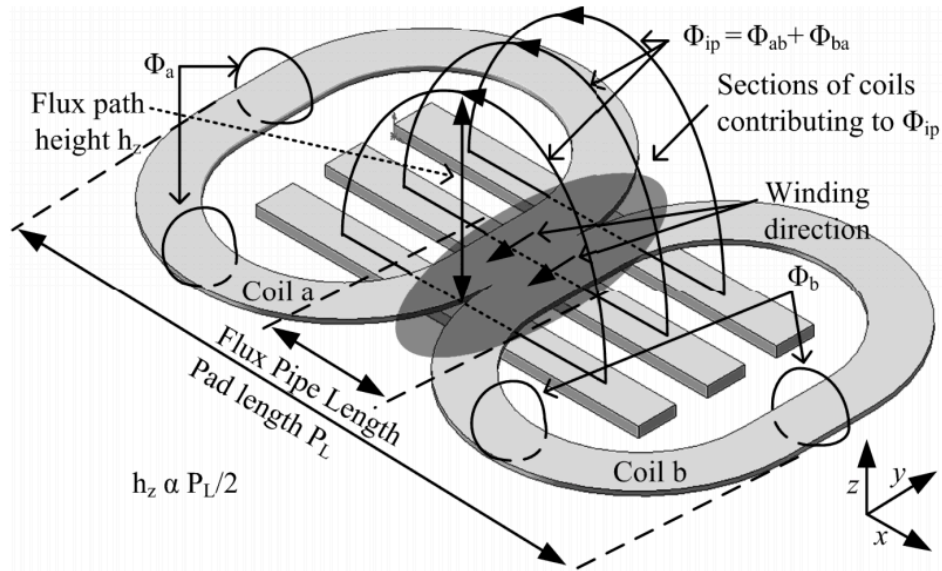
Flux extends to 1/4 of diameter for circular PAD, and to 1/2 of the length for rectangular PAD.



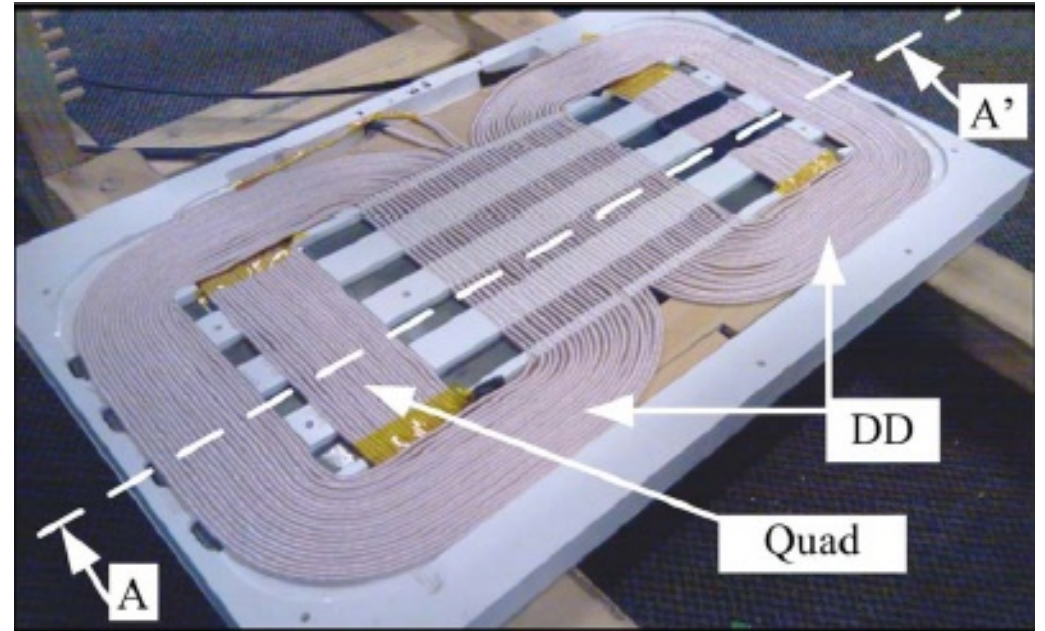
Pad Topology	(a)	(b)	(c)	(d)	(e)
P_{su} at given Separation (VA)	100mm 3841 150mm 1236 200mm 435	100mm 3832 150mm 1212 200mm 420	100mm 3479 150mm 1102 200mm 383	100mm 2825 150mm 868 200mm 293	100mm 2480 150mm 775 200mm 264
No. of bars	32.5	31.5	33	21	18
Use (VA/cm ³)	3.34	3.44	2.98	3.80	3.89



DD-DDQ



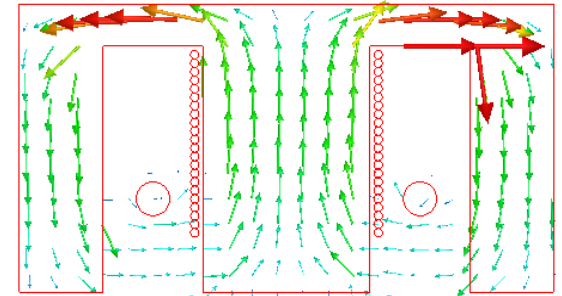
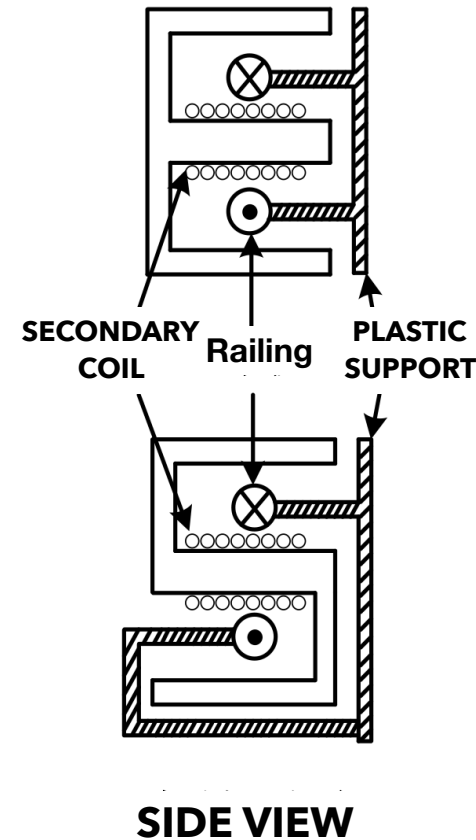
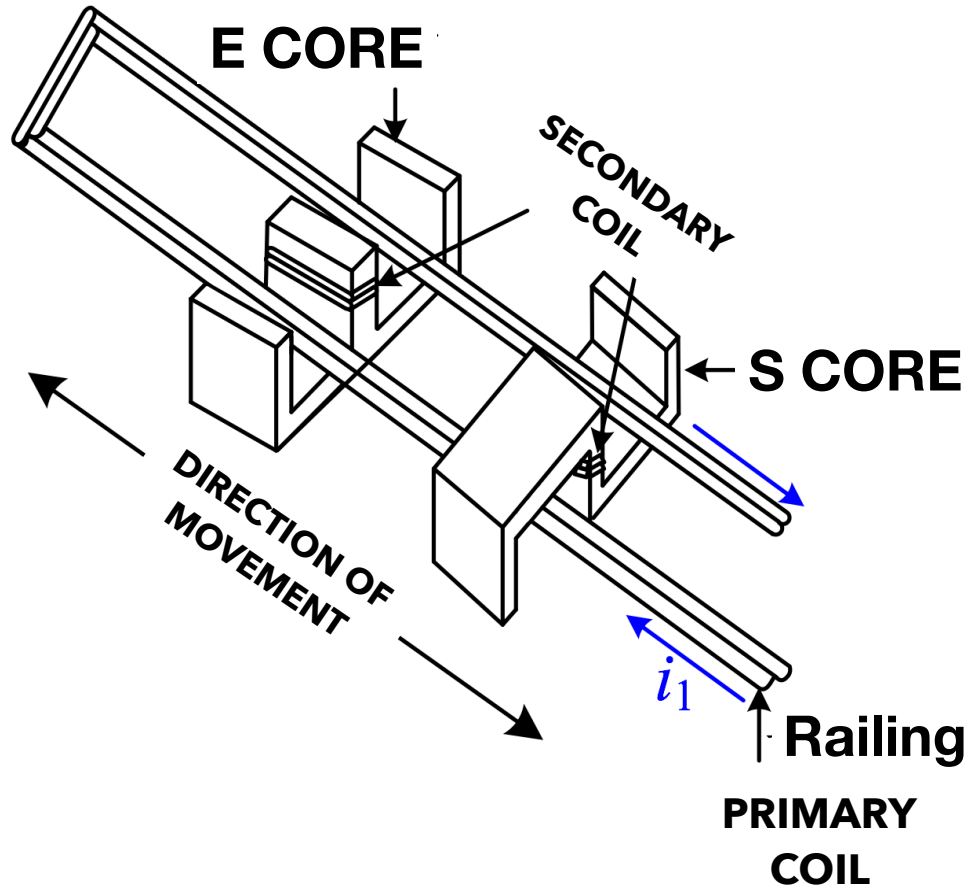
DD at primary



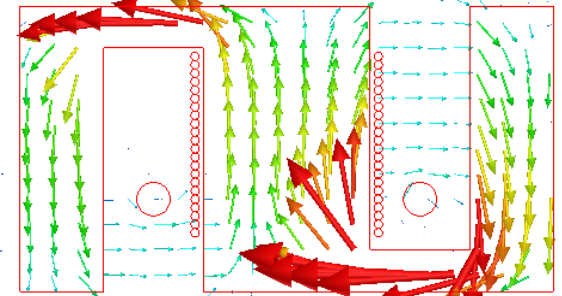
DD+Q at secondary

Basically the quadrature coil captures the flux at the null of the DD pad

Wireless Power Transfer in Trams

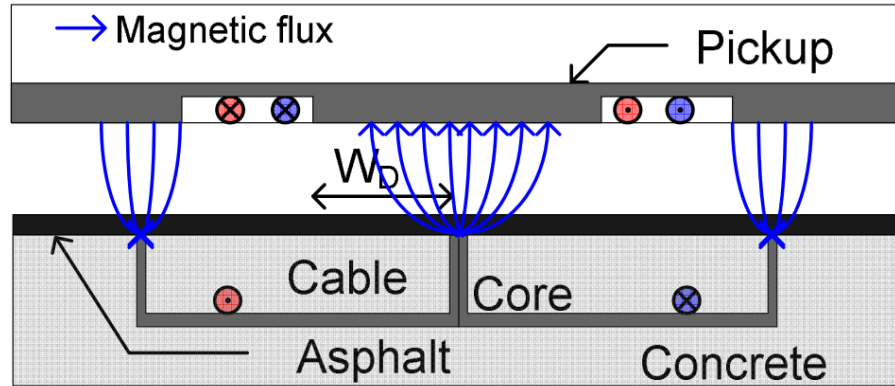


Fluxes partially cancel and less power transfer capability

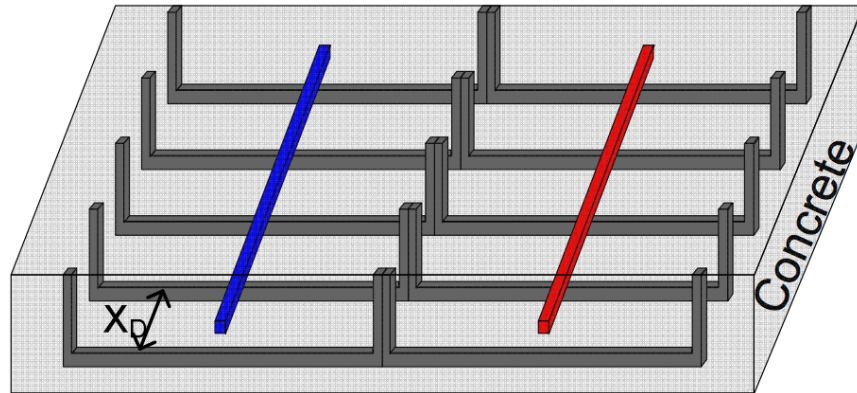


Less flux cancelled at the two sides and more power transfer capability

KAIST Bone Structure Core



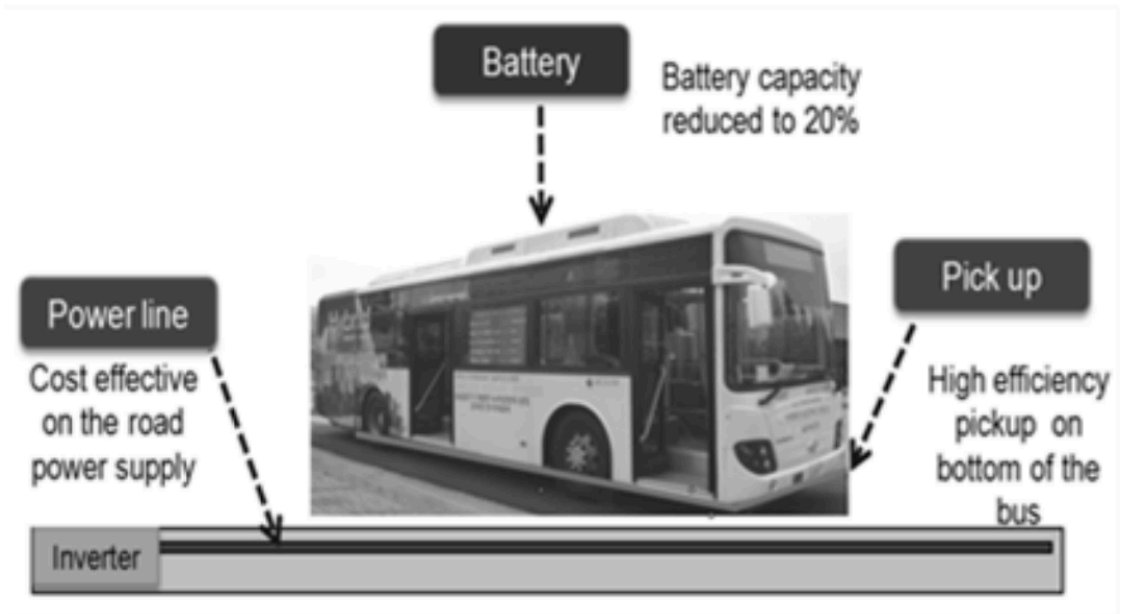
(a)



(b)

Amount of core significantly reduced.

Reduce EMF at center of road to below 60 mG to alleviate safety concern.



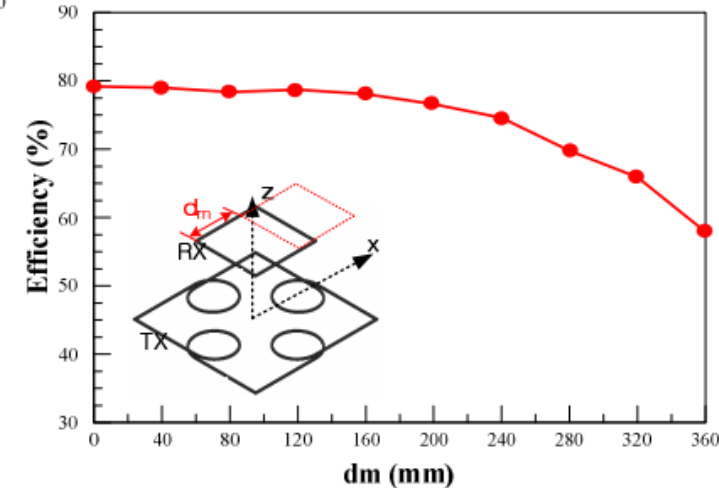
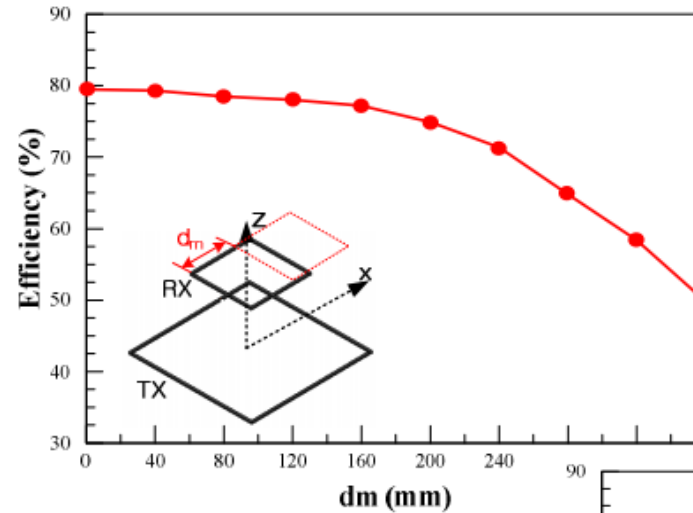
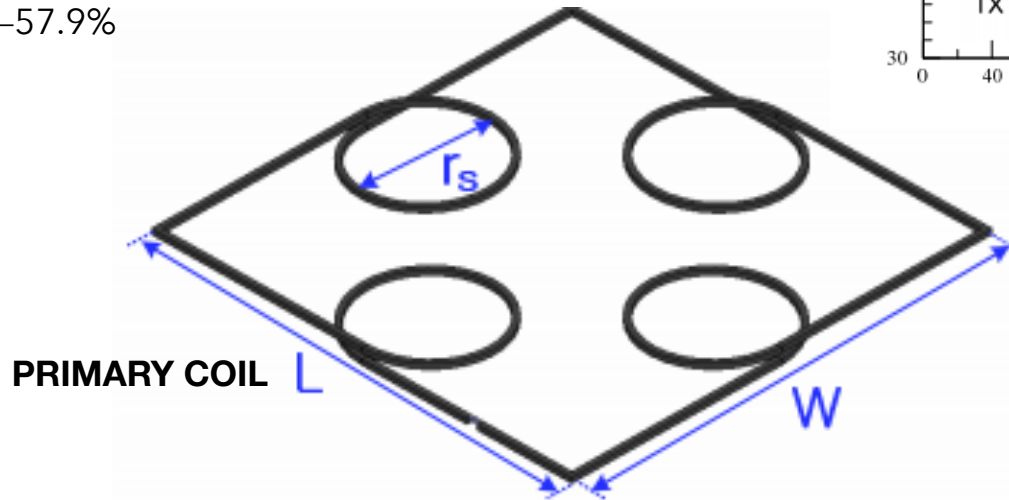
Coil Arrangement Enhancement

Small circular loops added on primary side.

Experiment:

Primary coil 800 mm x 800 mm; secondary coil 400mm x 400mm; gap 30 mm; misalignment 0–360 mm; small loops 200 mm diameter

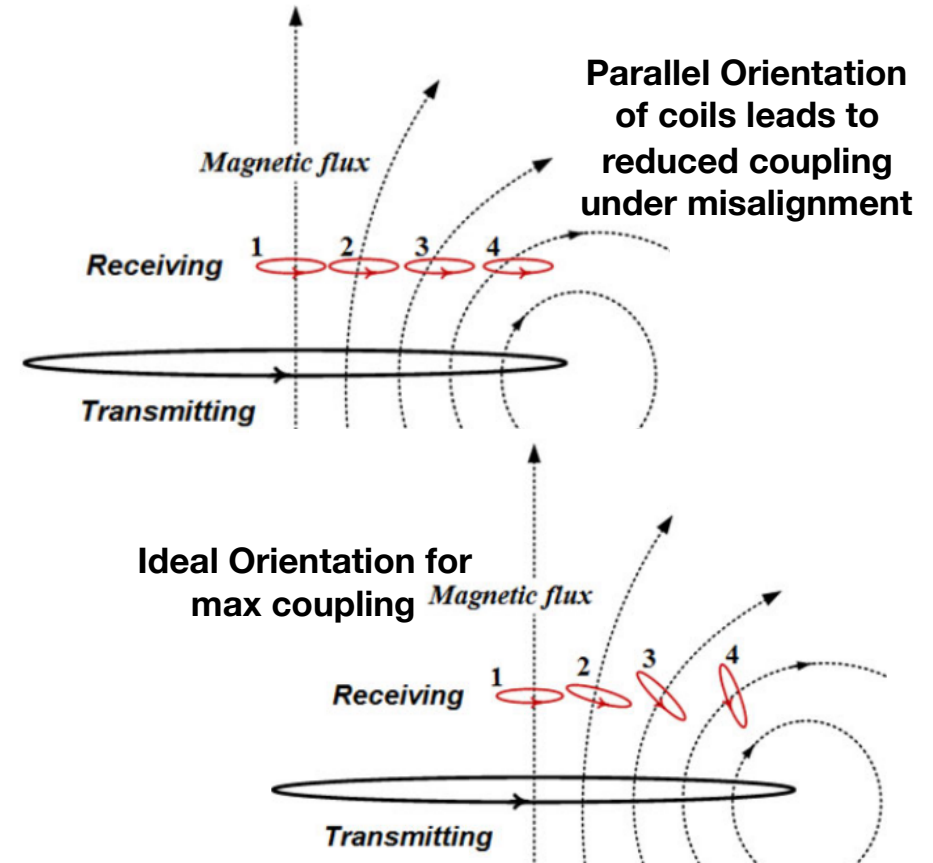
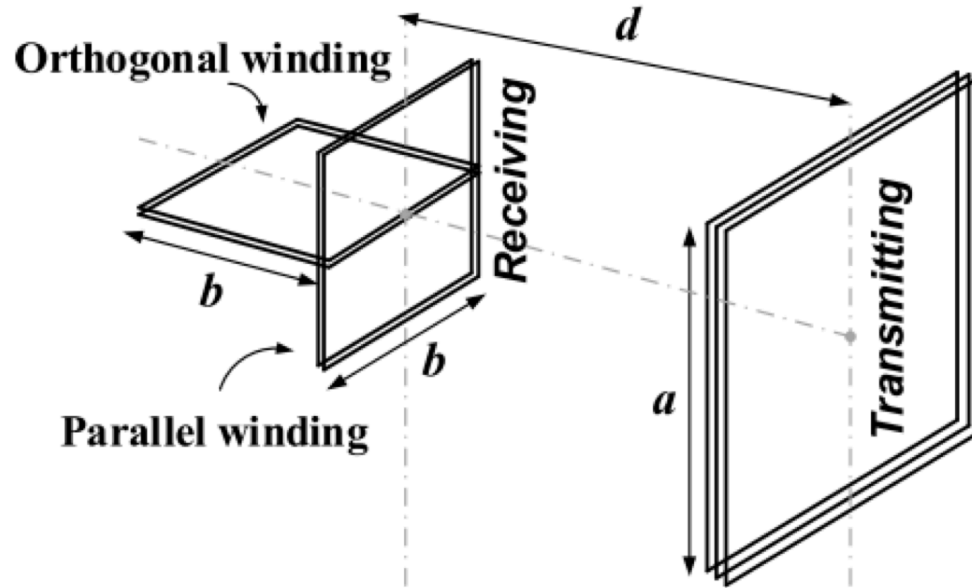
Efficiency improved from 79,5%–47,8% to 79%–57.9%



Reshaping Magnetic Fields

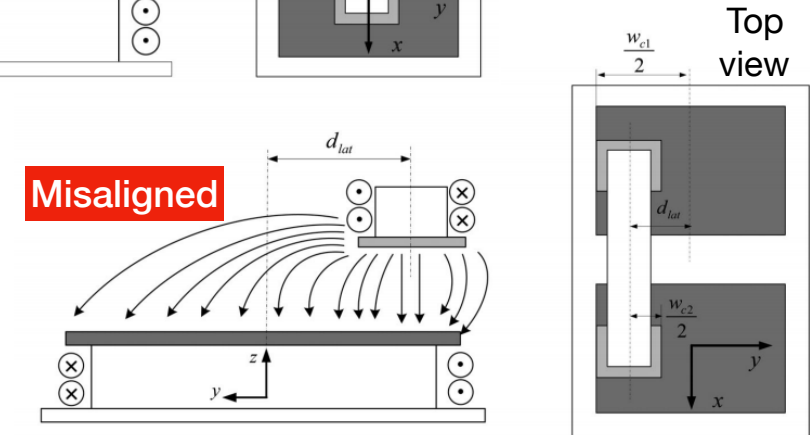
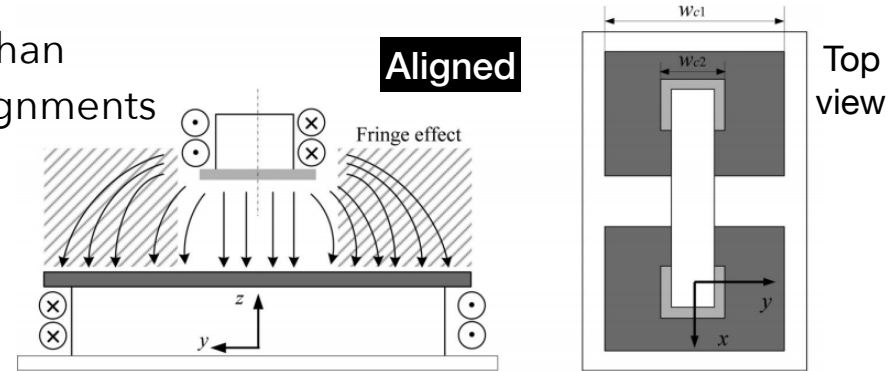
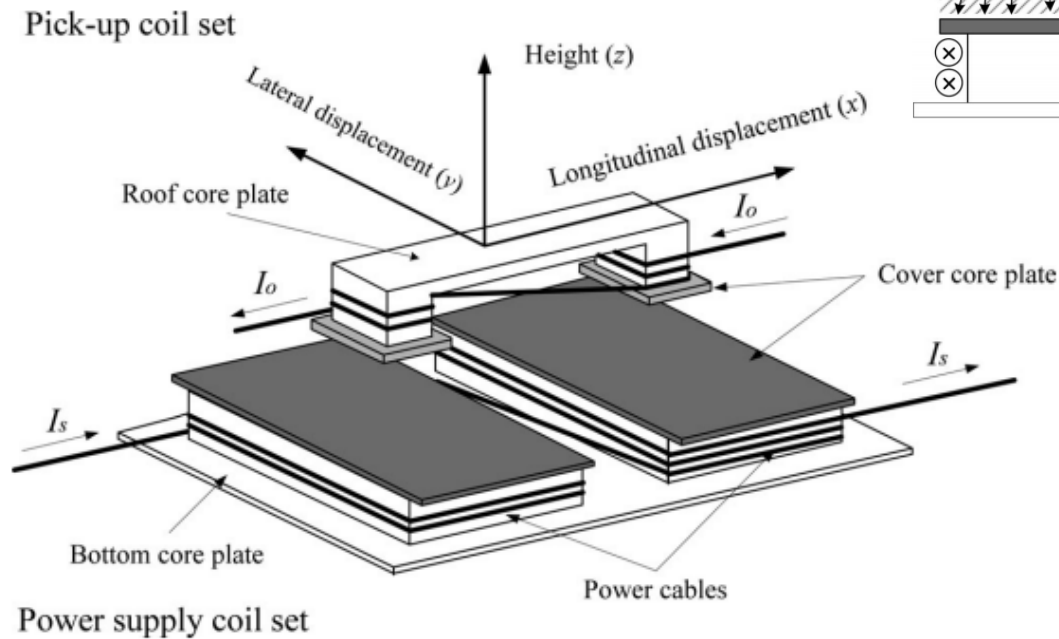
Two orthogonal coils in secondary

The pickup can be done in two directions, each working for one orientation and hence improving coupling under angular misalignment.



Asymmetric Design

Asymmetric design. Pickup coils much smaller than transmitting coils. Minimize the effects of misalignments in both x and y directions as well as gap size.

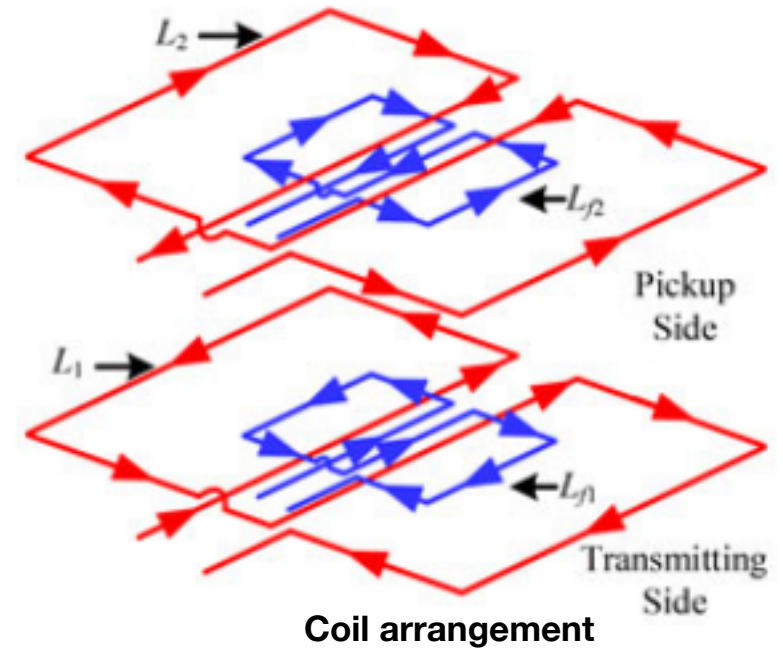
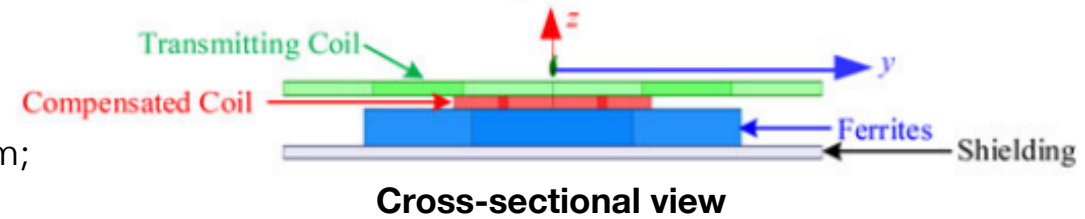
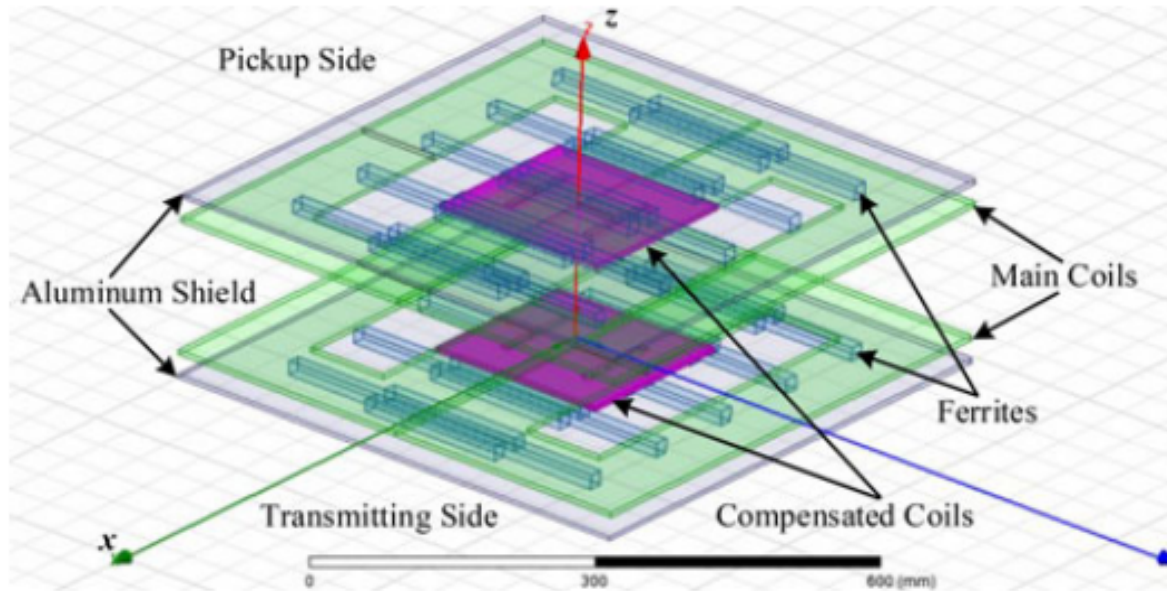


Compensation Coil Integrated Magnetics

LCC compensation with compensation coils wound on the same core as the DD pad.

Size 600 mm x 600 mm; compensation coil 200 mm x 200 mm;

Gap 150 mm; frequency 95 kHz; max efficiency 95.36%

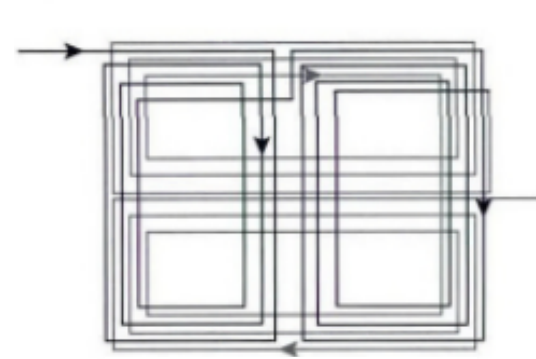
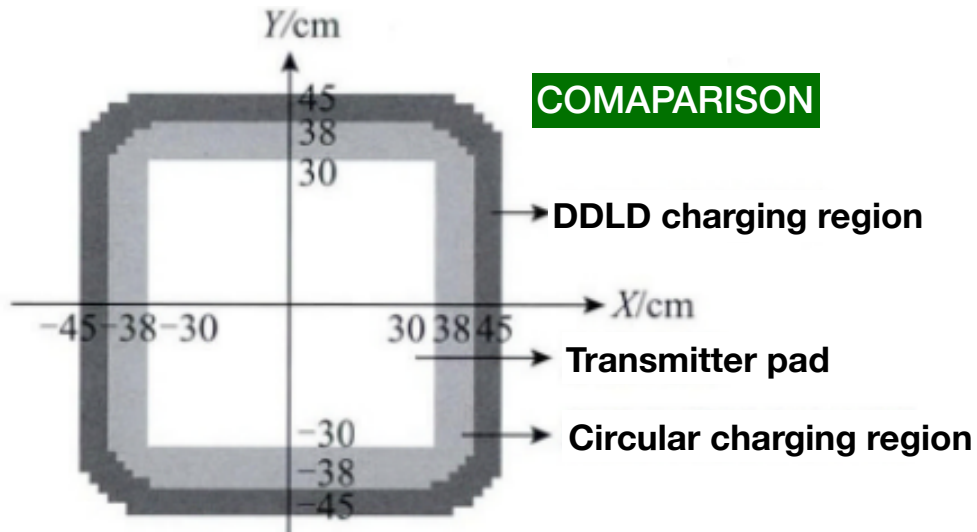


DDLD Charging Pad

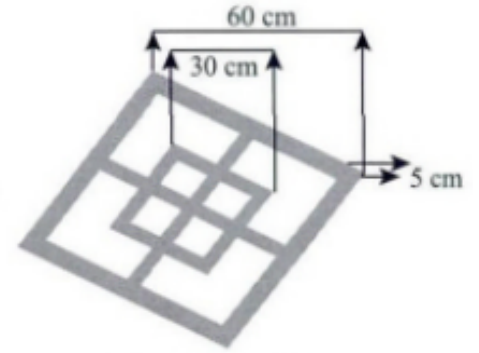
Chongqing University

DDLD design for magnetic coupling.

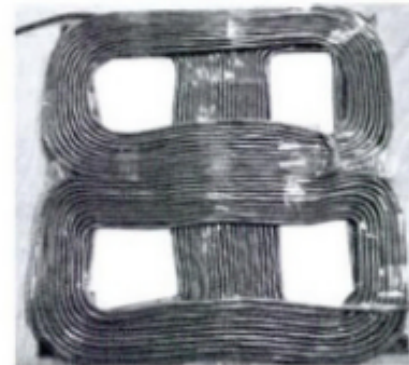
Wider alignment tolerance and wider charging gap.



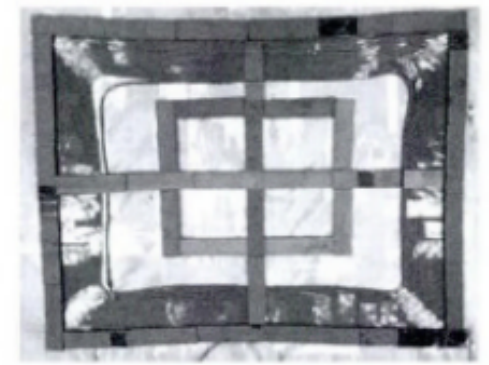
Winding arrangement



Ferrite design



Transmitter coil



Receiver coil

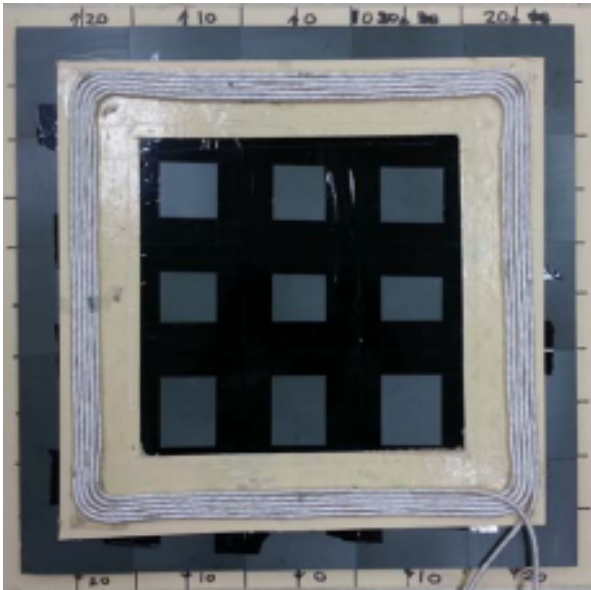
Add-on Coils

KAIST: Additional diagonal coils in primary side

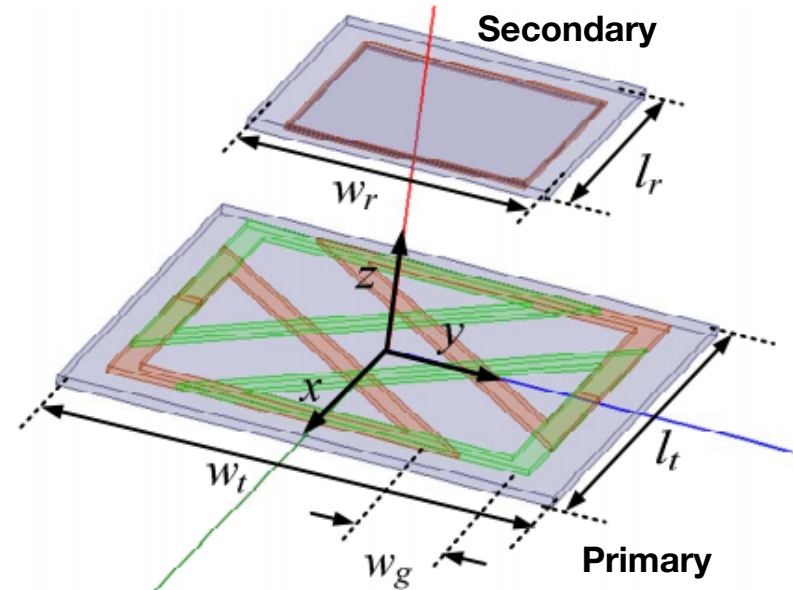
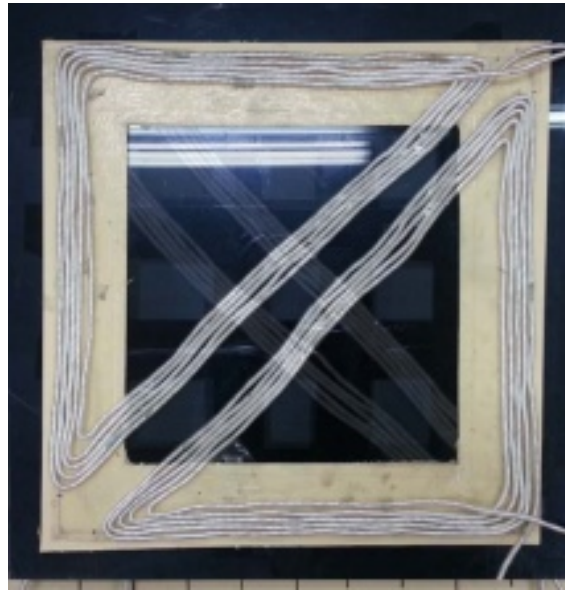
Added coils wound through diagonal lines

Improve misalignment along the diagonal directions

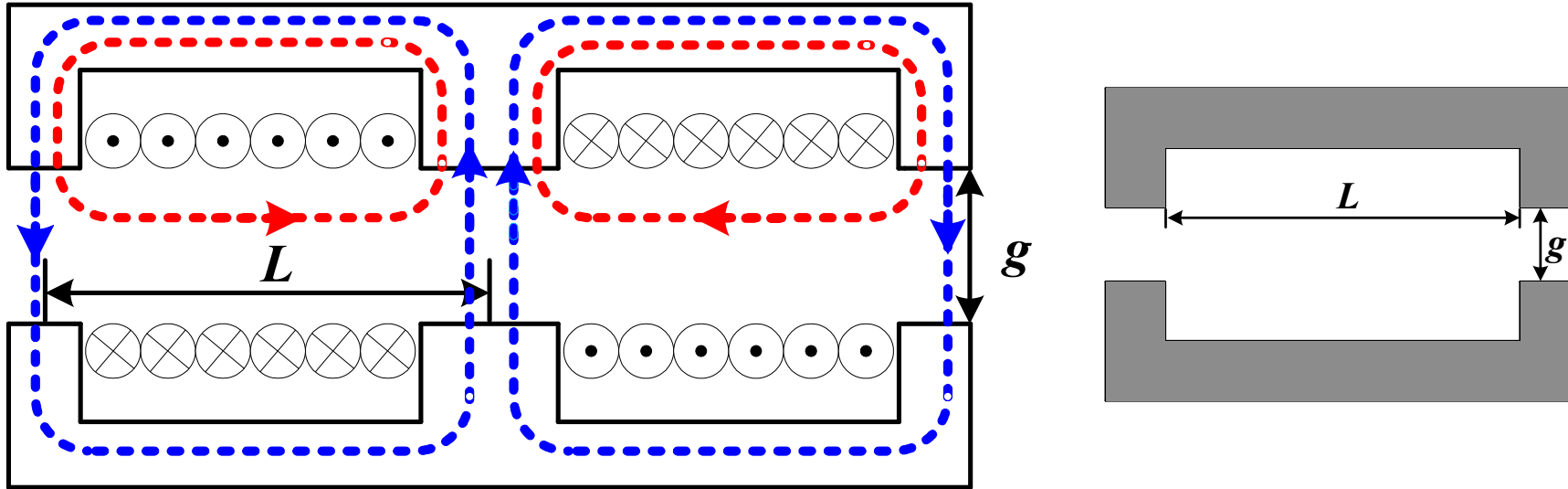
Traditional coils



DQ cross coils



Contactless Transformer Analysis

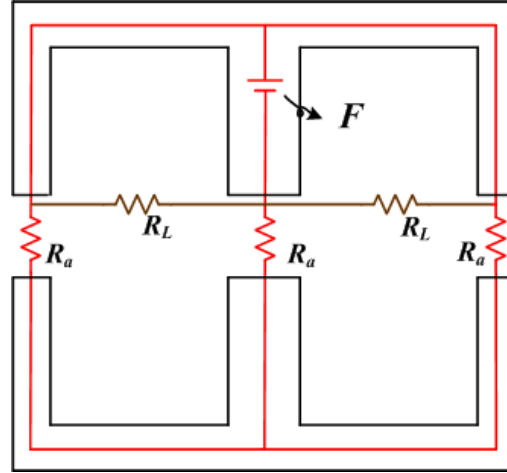
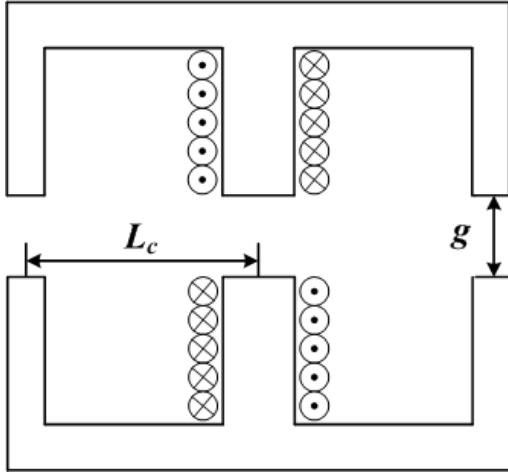


Coupling coefficient k depends on the ratio L/g .

Larger L gives larger k .

Smaller g gives larger k .

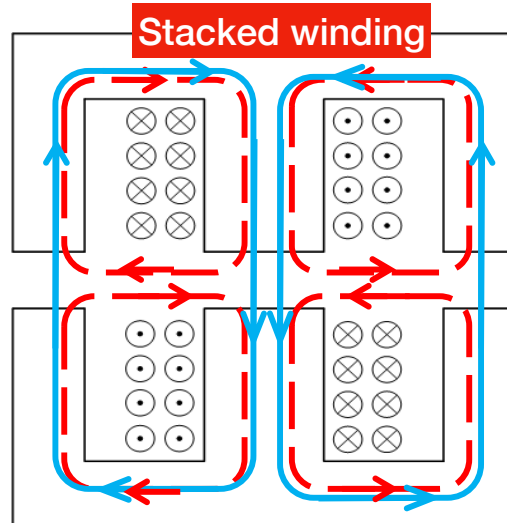
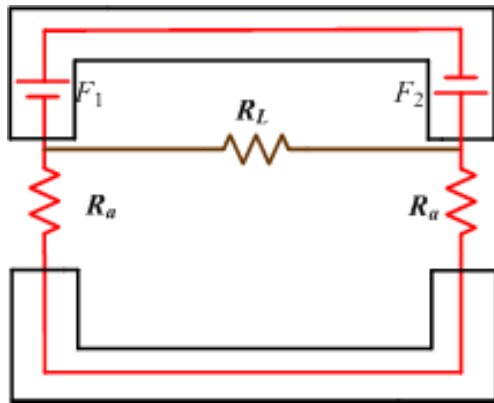
Nishimura Formula (Conventional reluctance model)



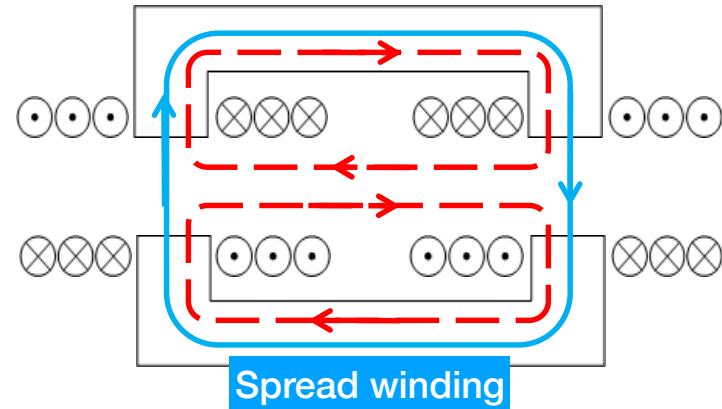
$$F = F_1 + F_2$$

$$k = \frac{(F_1 + F_2)/2R_a}{(F_1 + F_2)/2R_a + (F_1 + F_2)/R_L} = \frac{1}{(2g/L_c) + 1}$$

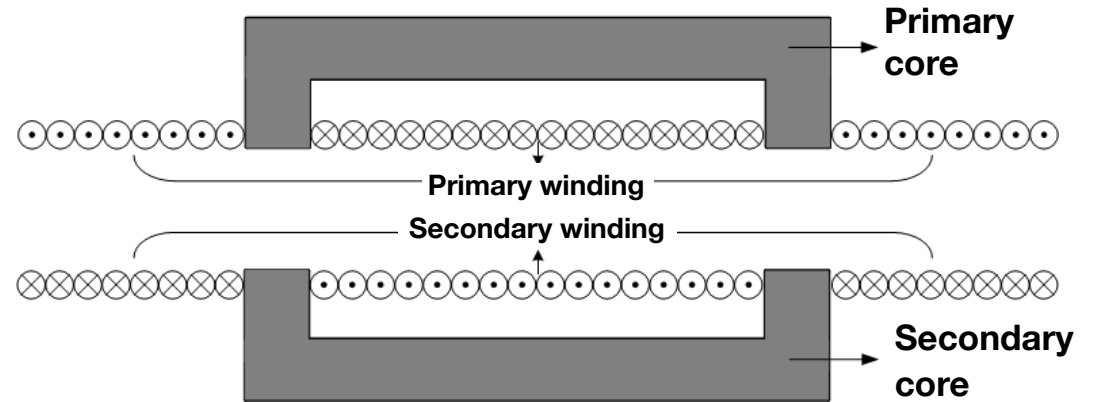
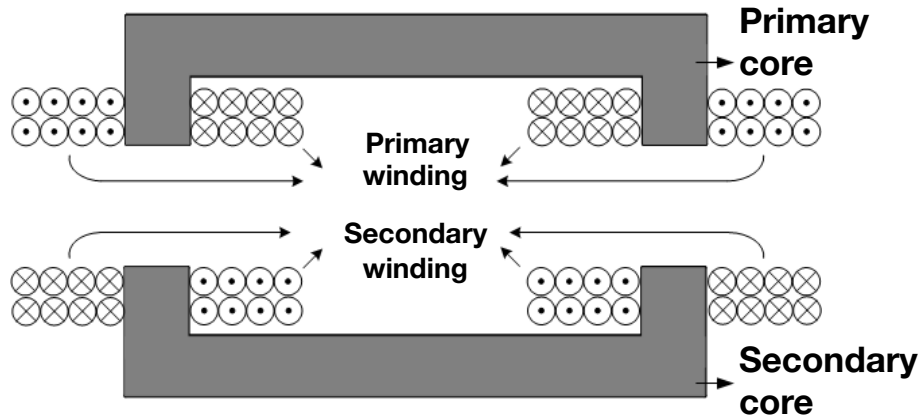
Thus, k is affected by the core and the winding arrangement.



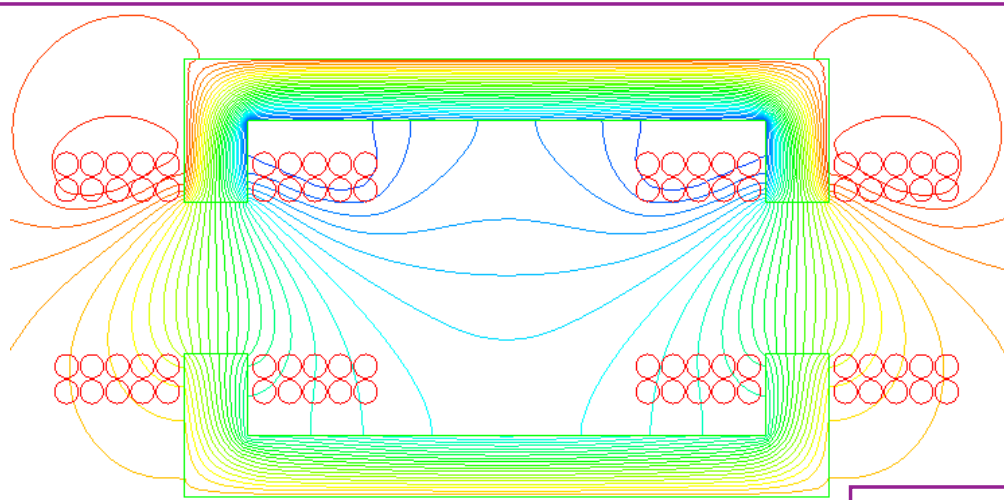
Changing the shape of the core, widening L



Winding: Stacked vs Spread



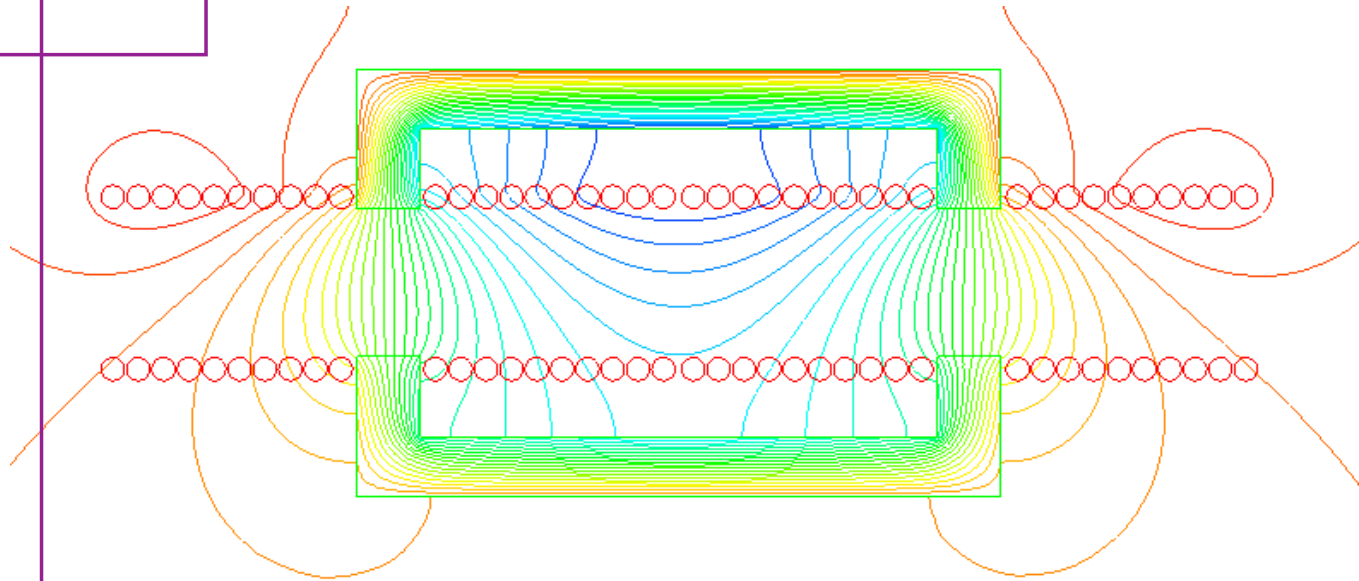
WINDING	CORE	Core Weight (gram)	L (mm)	Coupling Coeff.
Stacked	Planar U43 (Planar E with middle column removed)	59	34.7	0.39
Spread				0.46
Experimental Conditions : Both primary and secondary coils are 25 turns; gap; frequency 300 kHz				



STACKED WINDING

The stacked coil suffers the clouding effect reducing the flux linking through the gap. Some flux does not link to the secondary.

SPREAD WINDING has better coupling



Limitations

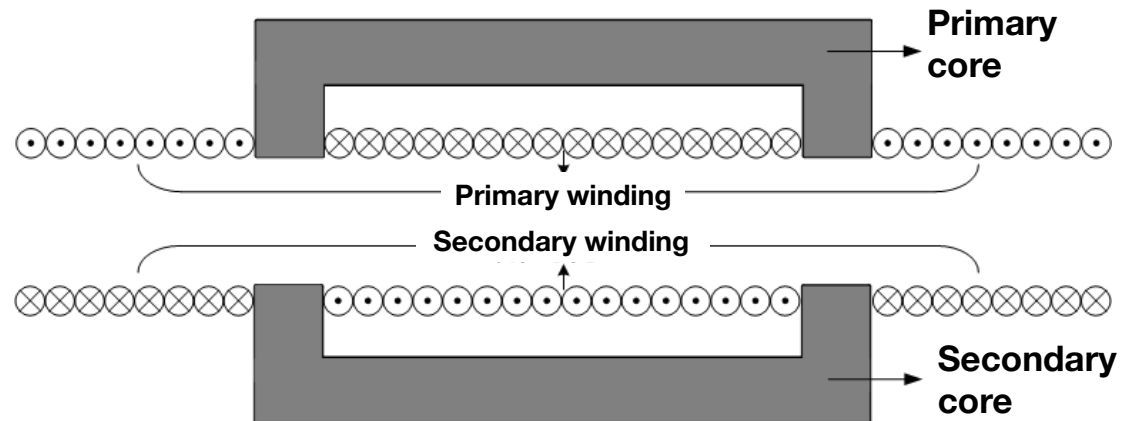
Nishimura formula cannot give consistent estimates of coupling coefficient for different experimental settings.

Experiment 1: Comparing planar core U43 with $L = 34.7$ mm with E64 core with $L = 21.8$ mm, same gap, the U43 has lower k !! Larger L gives smaller k !

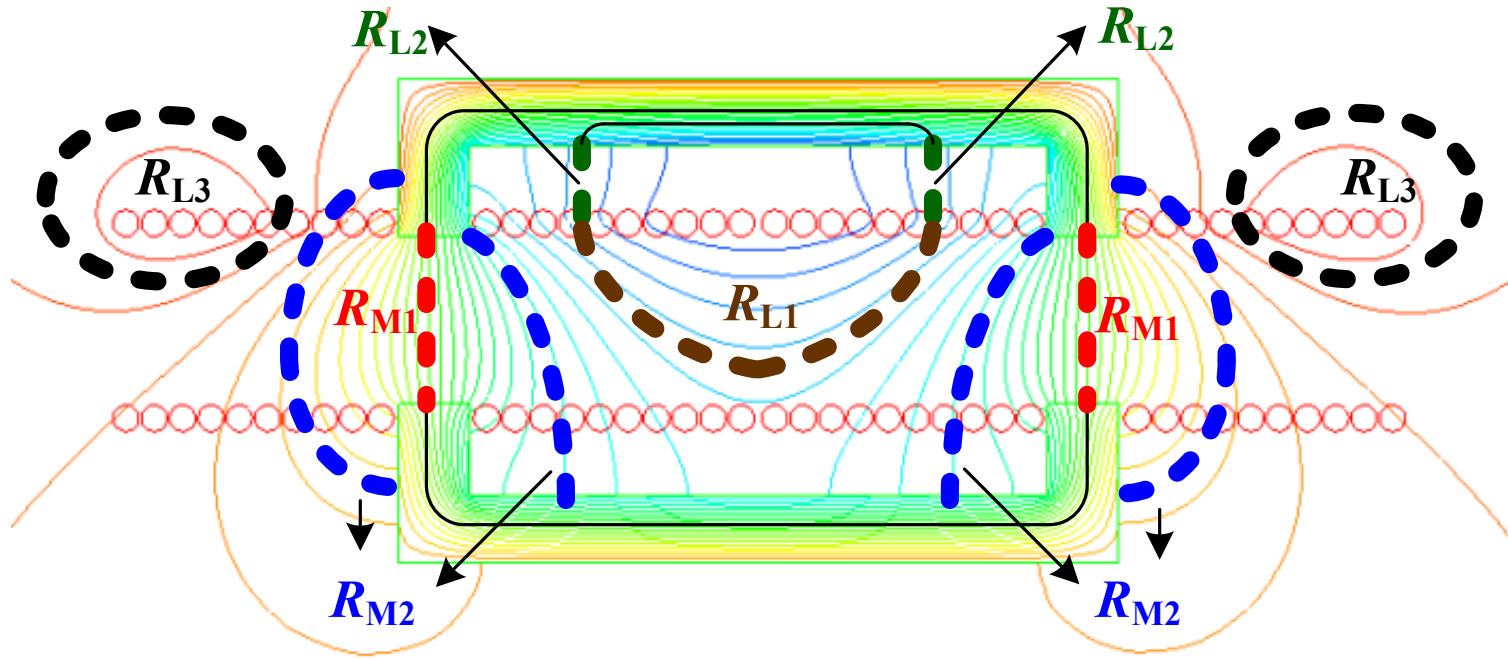
Experiment 2: Comparing U64 core (E64/10/50 without middle column) with $L = 34.7$ mm, with U43 core with $L = 53.8$ mm, gap at 10 mm, the coupling k changes only from 0.46 to 0.48. Not consistent with the formula. The increased L gives too little increase in k !

However, the formula gives $k = 0.634$ for U43, and $k = 0.64$ for U64, which are much bigger than experimentally measured values.

The Nishimura formular over-estimates k .



Optimization by magnetic path consideration

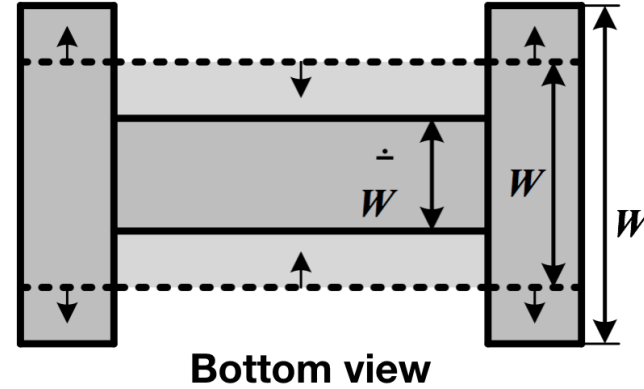
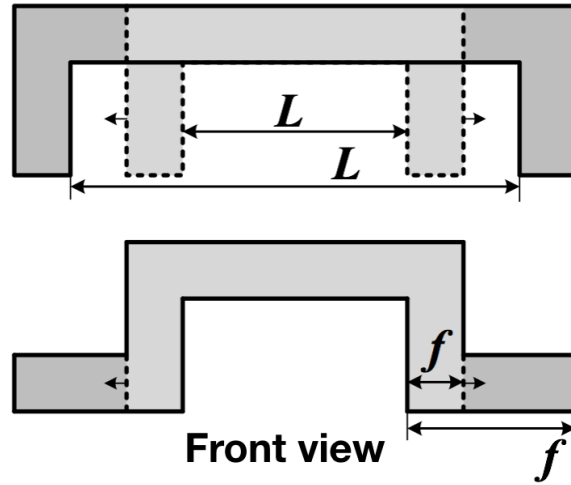


Reduce R_{M1} and R_{M2} , increase R_{L1} , R_{L2} and R_{L3} .

Optimization/Enhancement Method

Increase L so as
to reduce R_{M2}

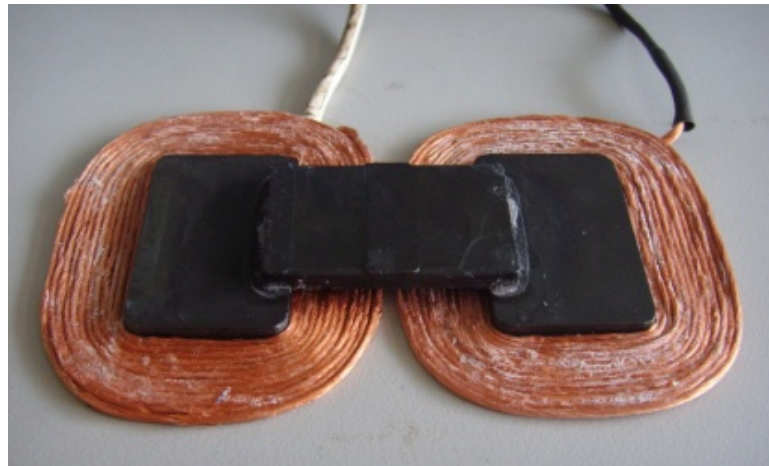
Increase f so as
to reduce R_{M1}



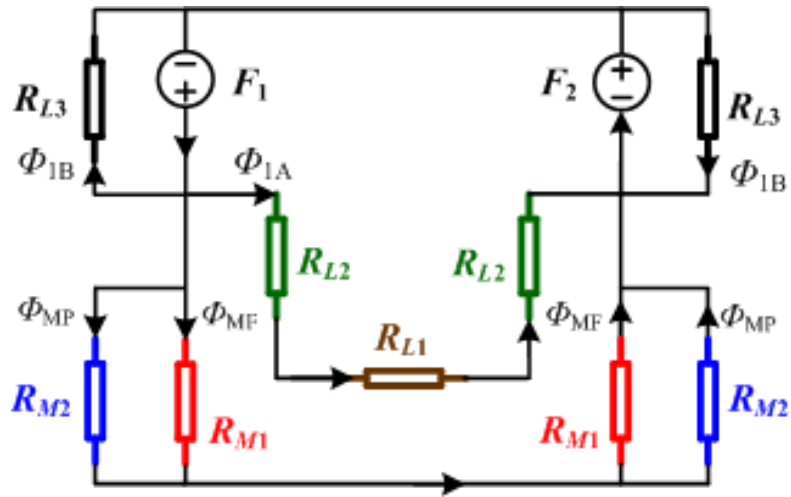
Increase W of
side columns so
as to reduce R_{M1}
and R_{M2}

Reduce W of
center column
so as to increase
 R_{L1} and R_{L2}

**Planar U core
Extended rectangular
coil**



Modeling of Contactless Transformer



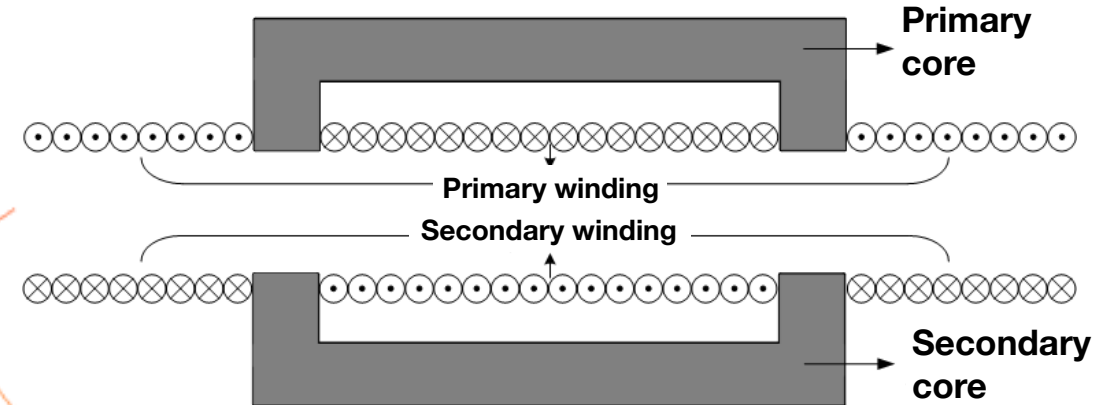
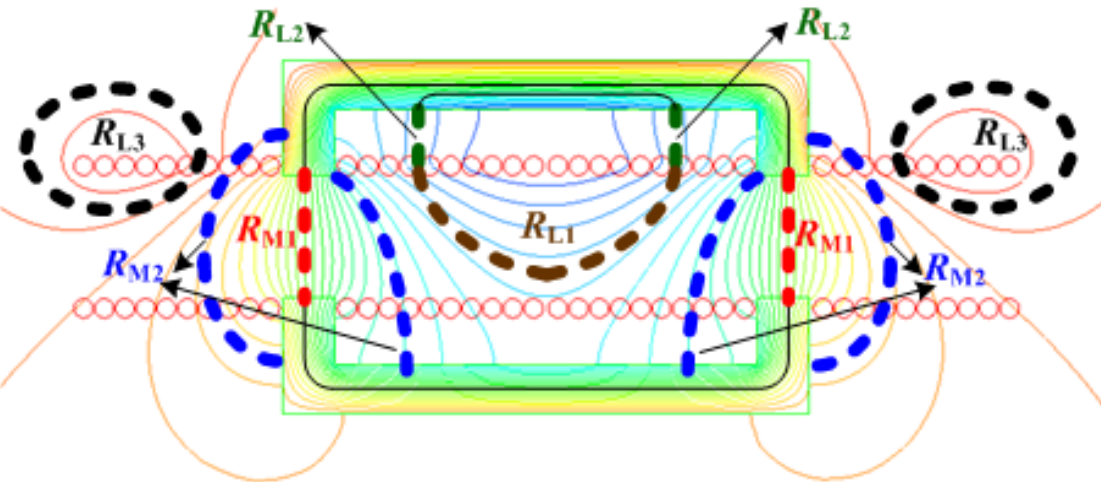
Magnetic path model

Leakage inductance path resistance: R_{L1} , R_{L2} and R_{L3}

Full coupled mutual inductance path resistance: R_{M1}

Partial coupled mutual inductance path resistance: R_{M2}

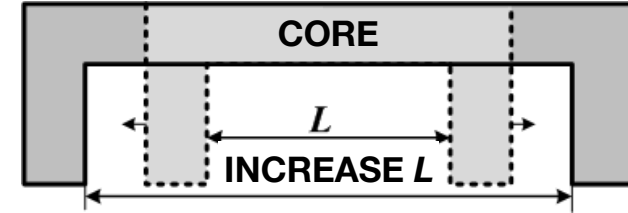
Full/partial coupling: Primary and secondary coils are linked fully/partially.



Enhancement of Core Structure

Traditional approach

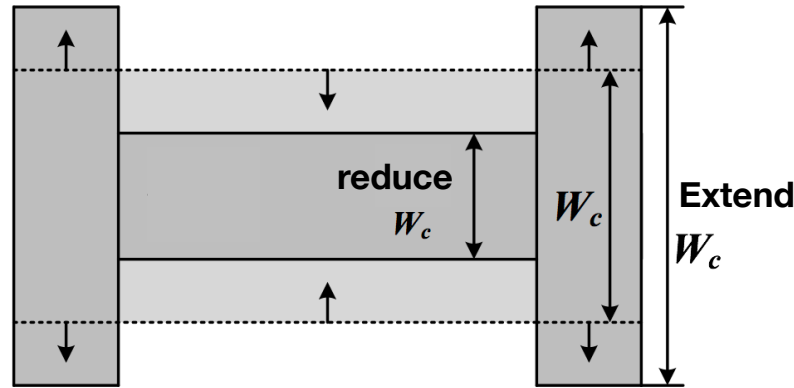
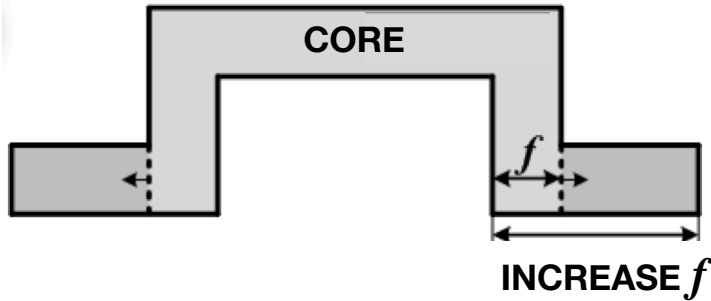
Increase coupled magnetic path; reduce leakage path



Enhancement from systematic analysis

Transformer core modification

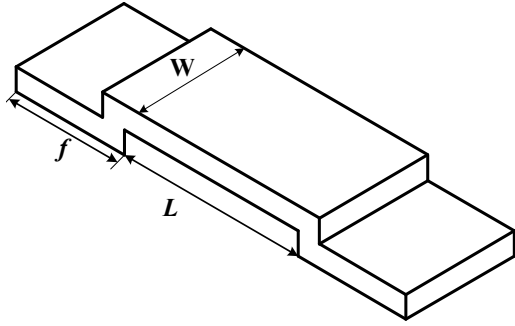
L is about twice the gap size



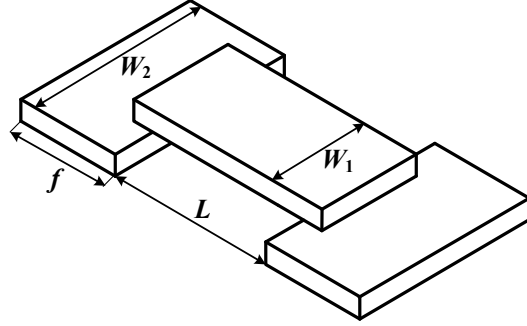
New core structure significantly enhances coupling

Extended Planar U-cores

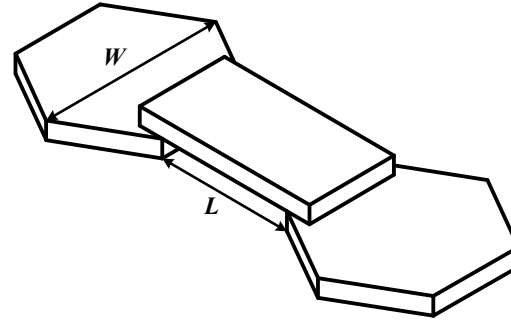
One-direction
extension



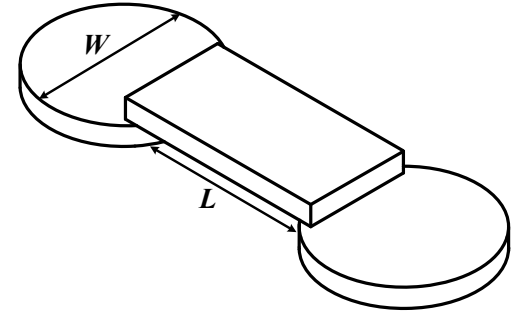
Two-direction
extension



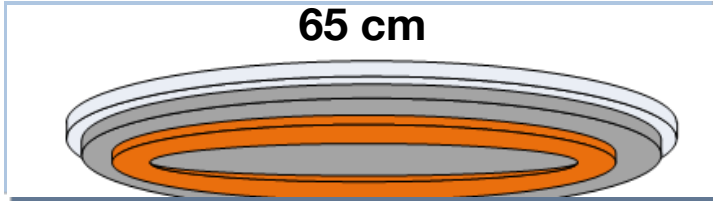
Polygon
extension



Circular
extension

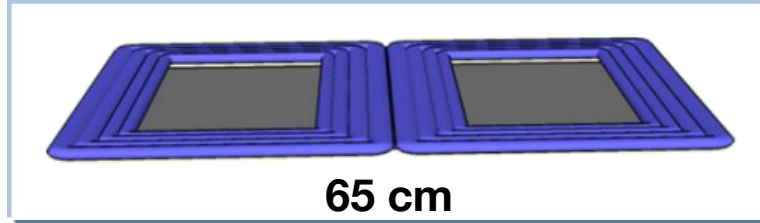
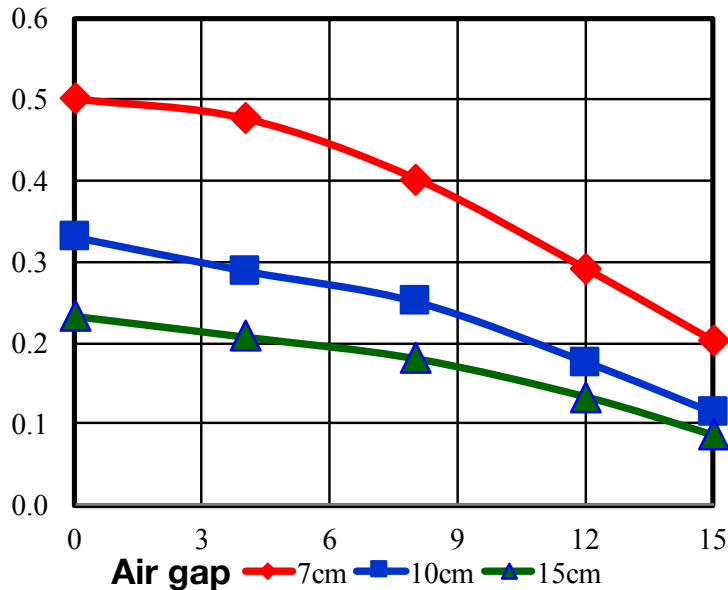


Misalignment



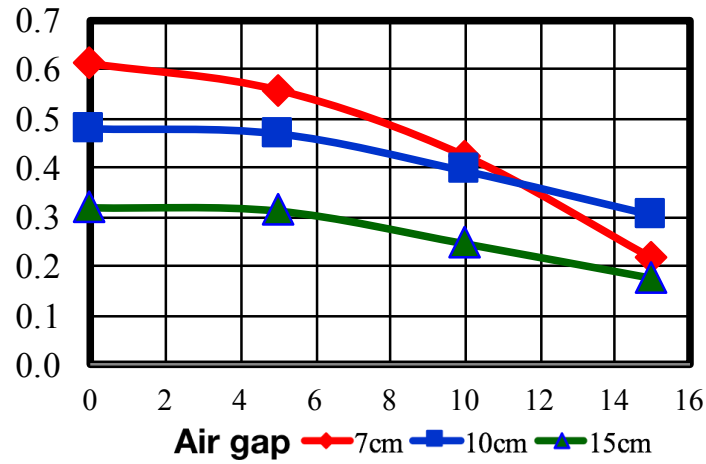
CIRCULAR PAD

coupling coeff vs. displacement



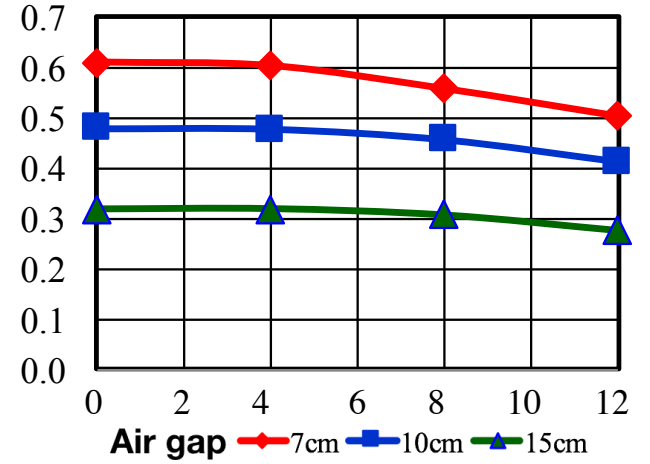
DD PAD

coupling coeff vs. x-displacement



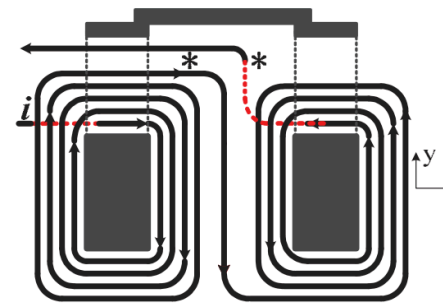
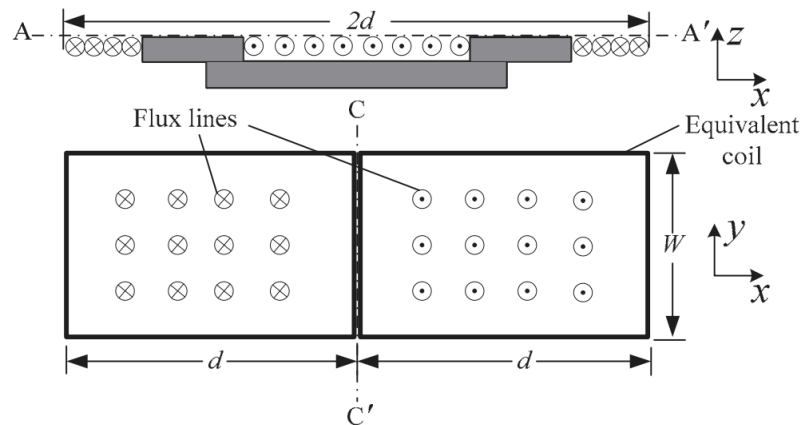
DD PAD

coupling coeff vs. y-displacement

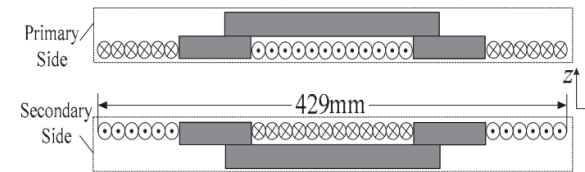


Detailed Models of DD Pads

(Ke *et al.*, IEEE ECCE 2016)



(a) Coil structure



(b) Length of the transformer

Key analysis:

Calculation of coupling coefficient

Null position (offset at which coupling is zero)

Model for misalignment

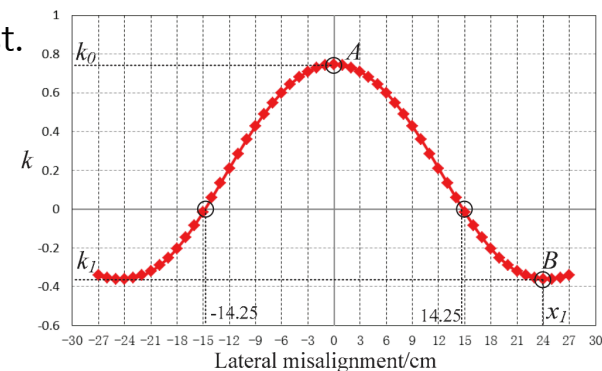
$$k(x) = a \sin(bx + c)e^{mx} + h \quad \text{by simulation}$$

where x = misalignment dist.

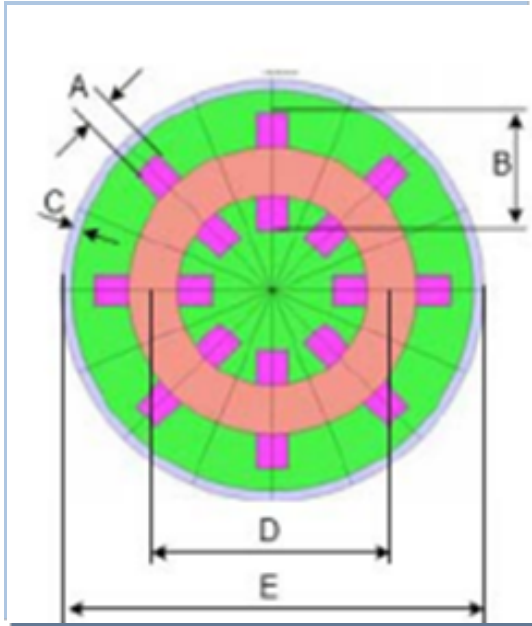
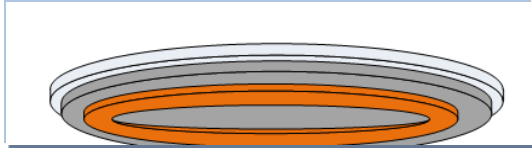
$$\Psi_x = BW(2d - 3x)$$

NULL position at

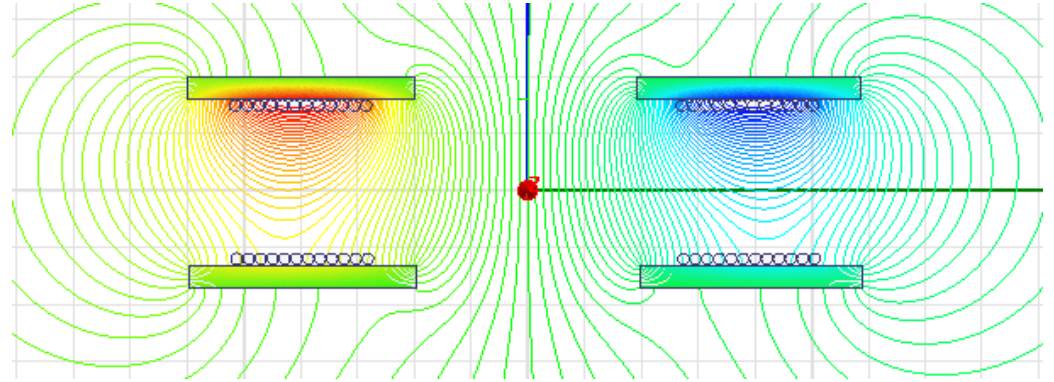
$$x = 2d/3$$



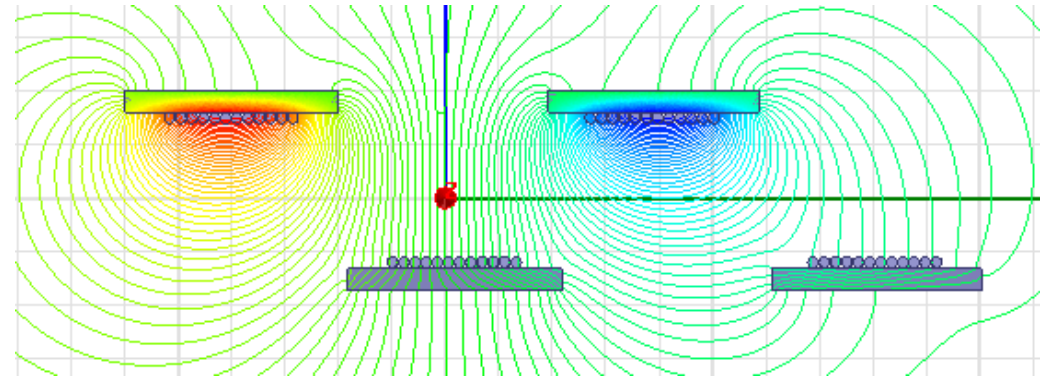
Circular Pad



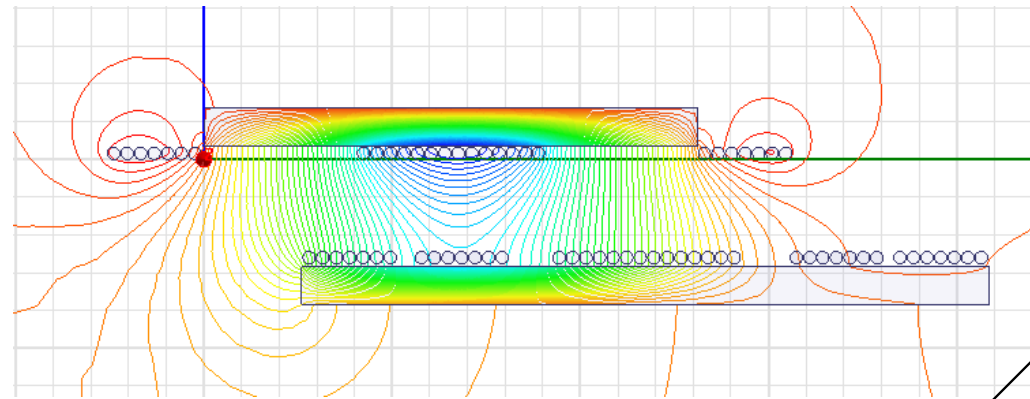
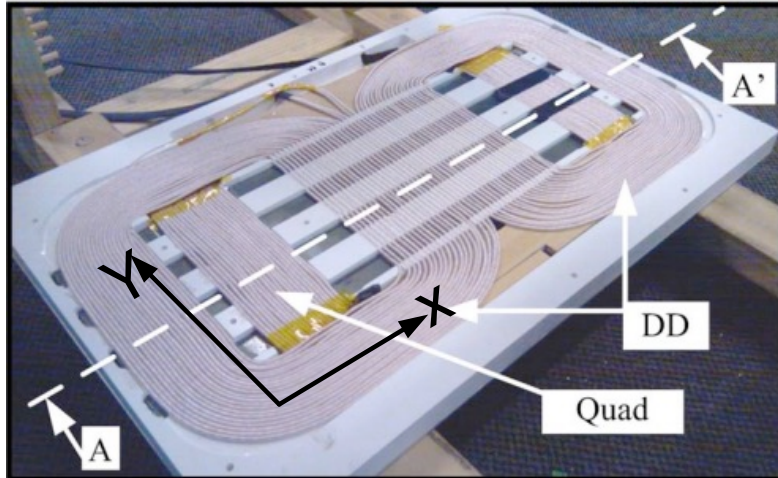
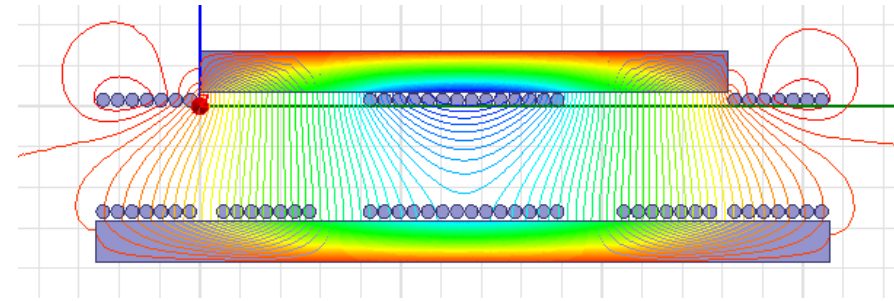
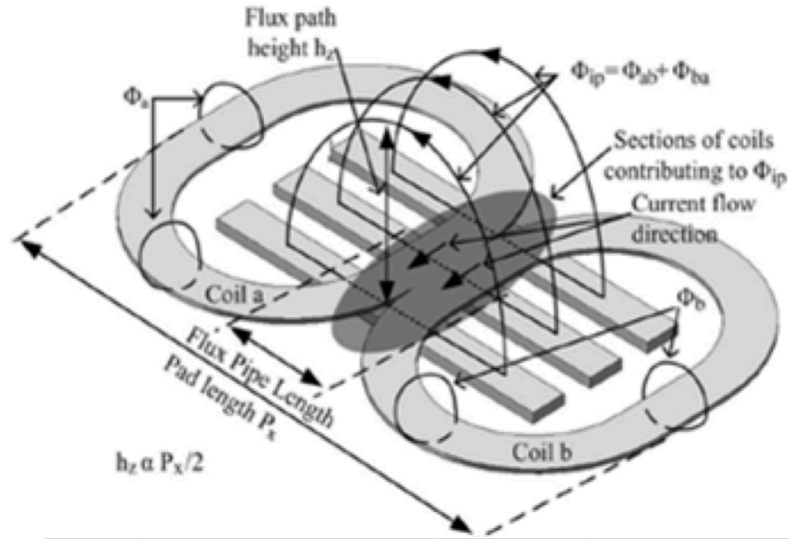
$$k = 0.515$$



$$k = 0.135$$

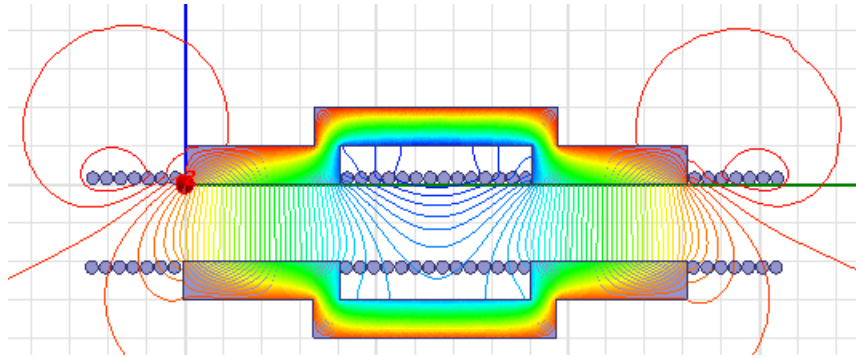
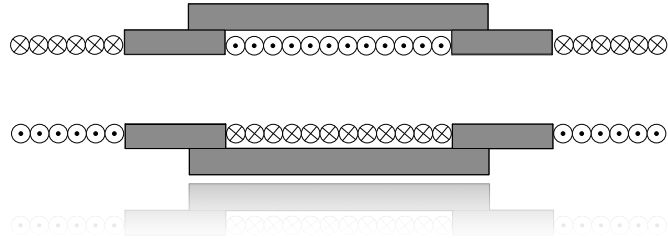


DDQ Pad

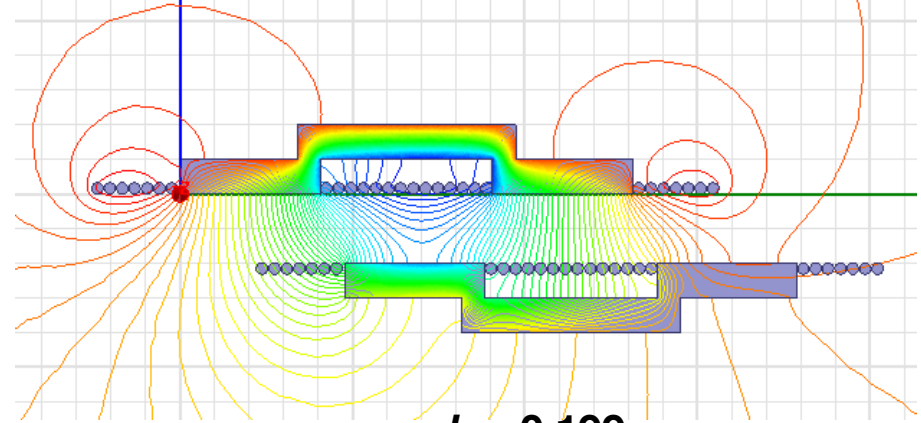


Quadrature coil links flux when misaligned

Extended U-core Transformer

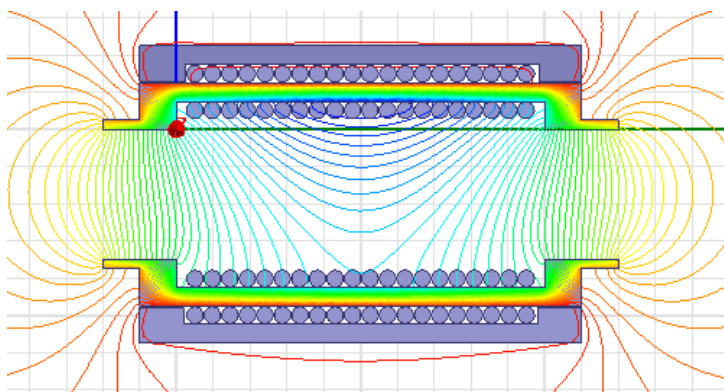
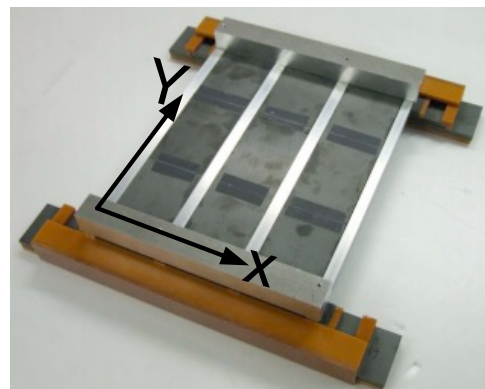


$k = 0.763$

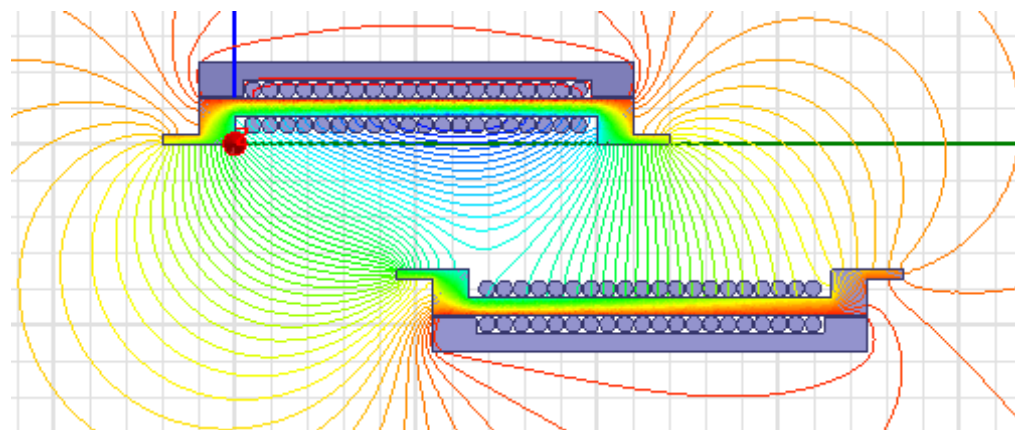


$k = 0.199$

Double-sided Core Transformer

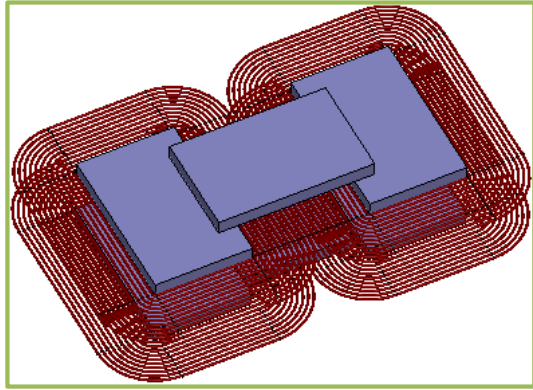


$$k = 0.402$$

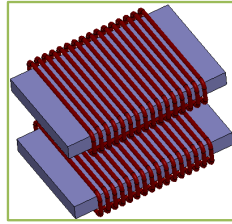


$$k = 0.179$$

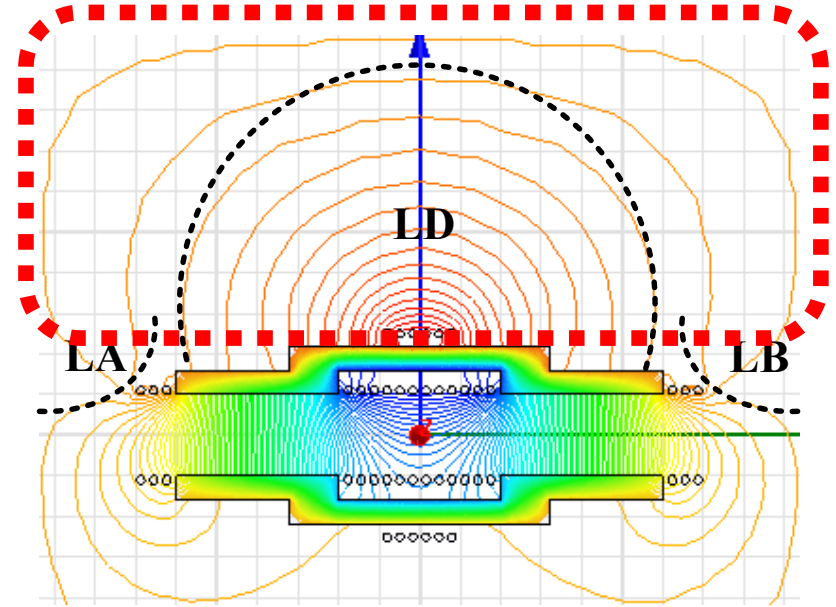
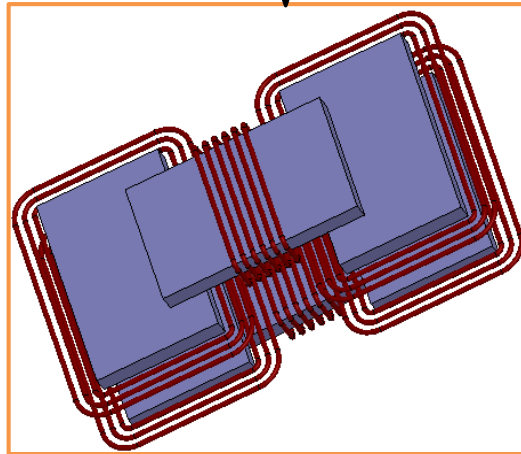
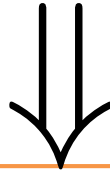
Additional Winding



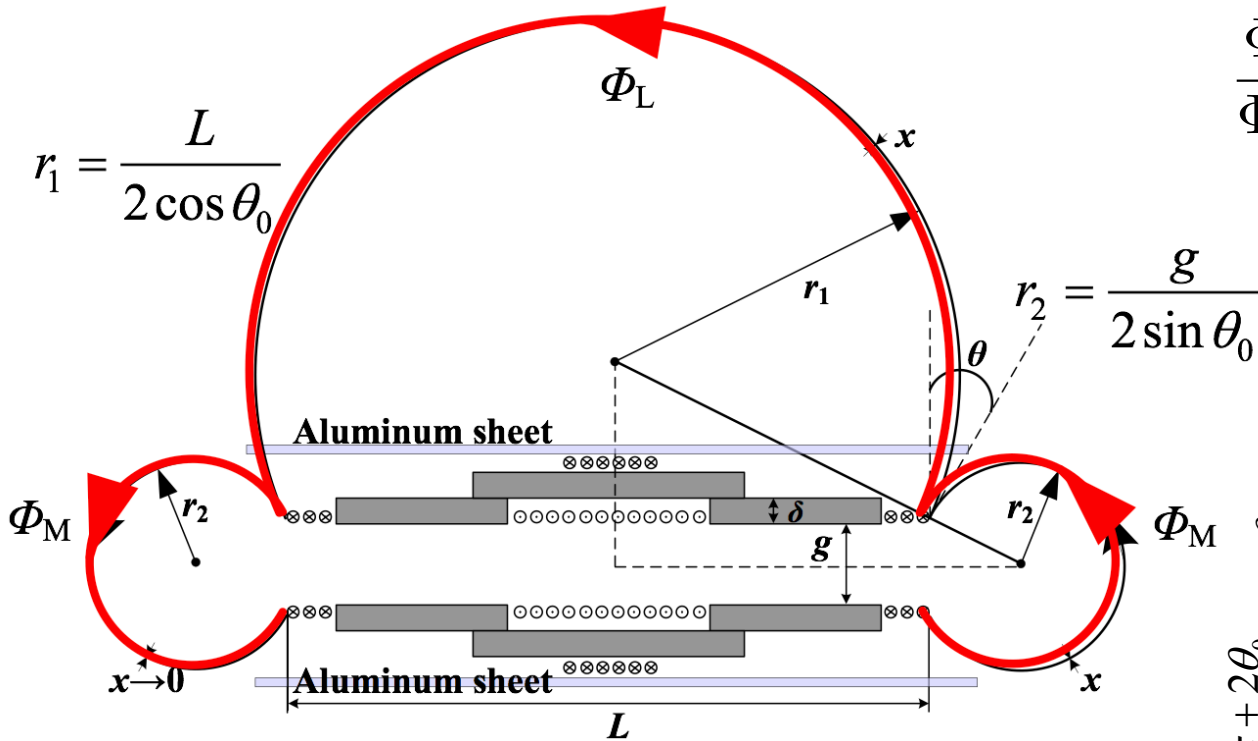
PLUS



Added winding



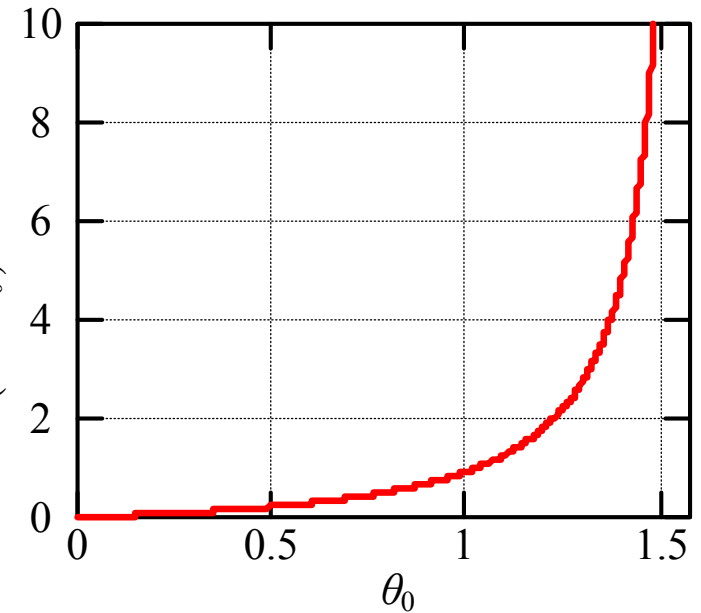
Shielding Enhancement



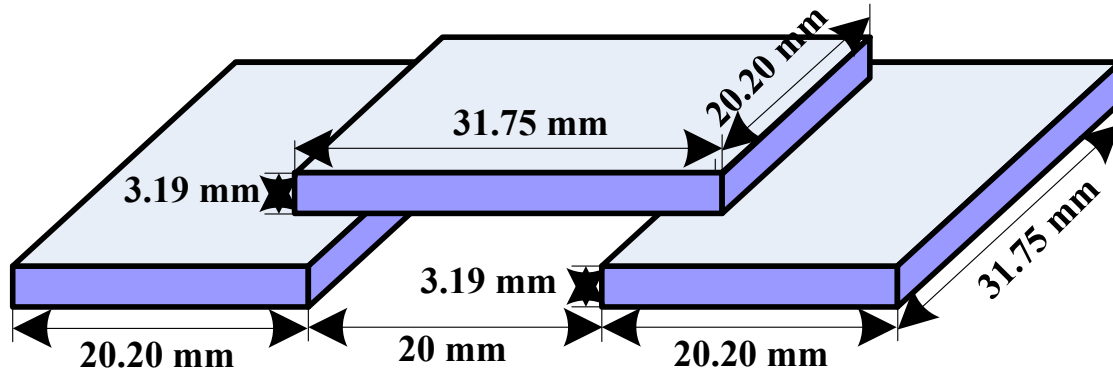
$$\frac{\Phi_{av_L}}{\Phi_{av_M}} = \frac{\pi + 2\theta_0}{4(\pi - \theta_0)} \tan \theta_0 \left(\frac{L}{g} \right) > 1$$

$$\frac{\pi + 2\theta_0}{(\pi - \theta_0)} \cdot \tan \theta_0 > \frac{4g}{L} \text{ is desired}$$

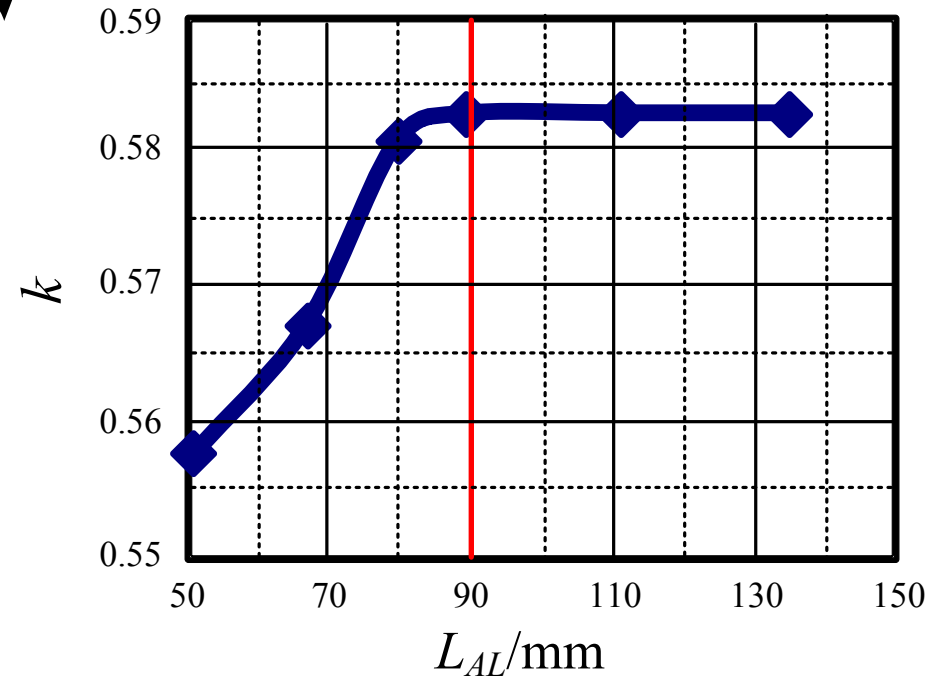
$$\frac{\pi + 2\theta_0}{(\pi - \theta_0)} \cdot \tan \theta_0$$



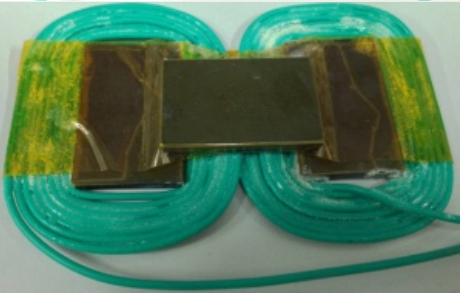
Experimental Results



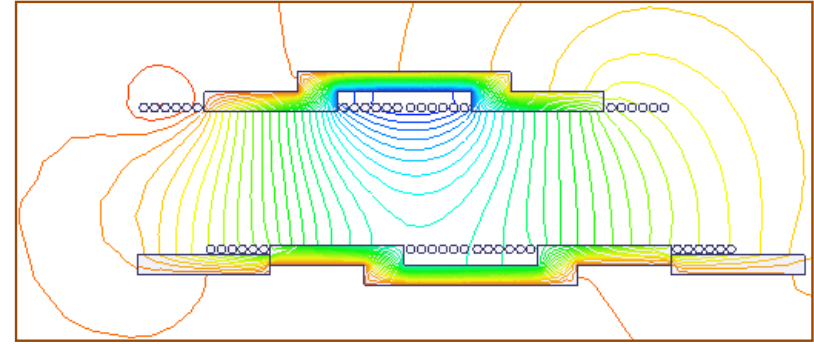
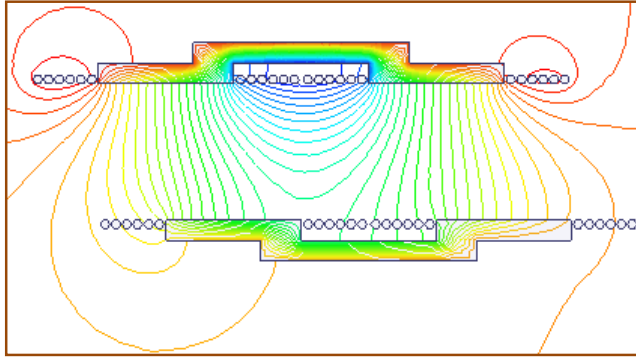
$$L_{A1} \geq L + 2(2\delta + d) \tan \theta_0$$



Experimental Results

	TRANSFORMER PHOTOS	L (mm)/ g (mm)	k
Vertical winding		$330/50=6.6$	0.48
Planar side winding		$90/10=9$	0.523
Combined winding		$73/10=7.3$	0.583

Extended U-core Misalignment

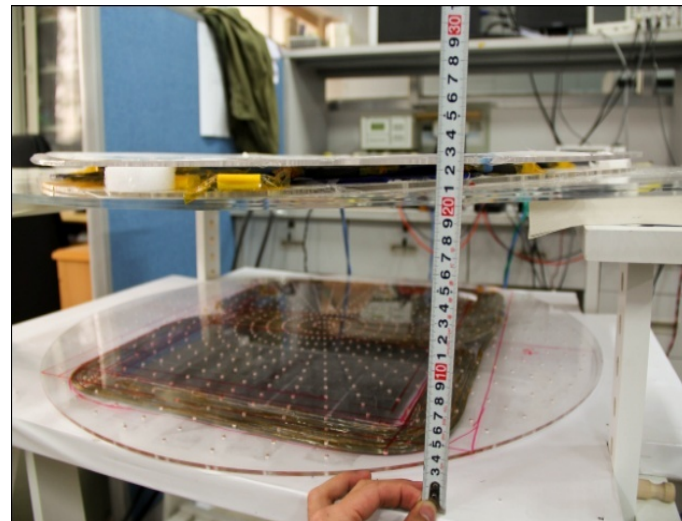
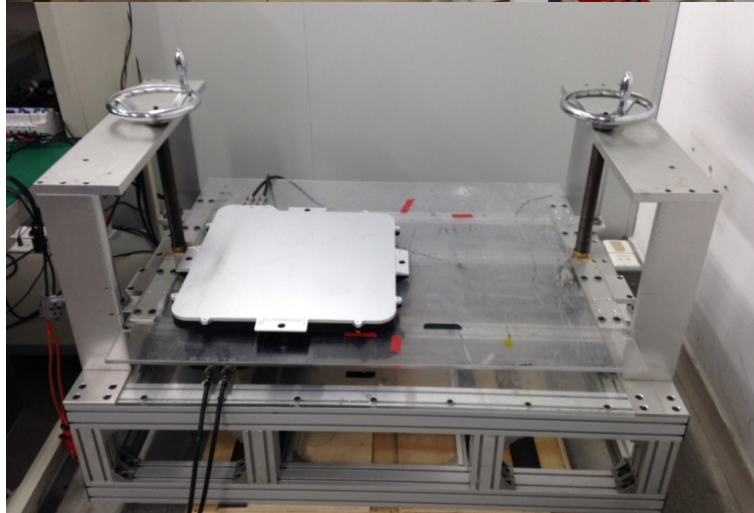


1kW System Efficiency	Gap 20 cm Perfectly aligned	Gap 26 cm Perfectly aligned	Gap 26 cm Displaced 14 cm
Traditional U-core	91.25%	87.59%	81.67%
Extended U-core with combined windings		88.14%	82.75%

NUAA Collaborator's Lab at Nanjing

Prof. Qianhong Chen

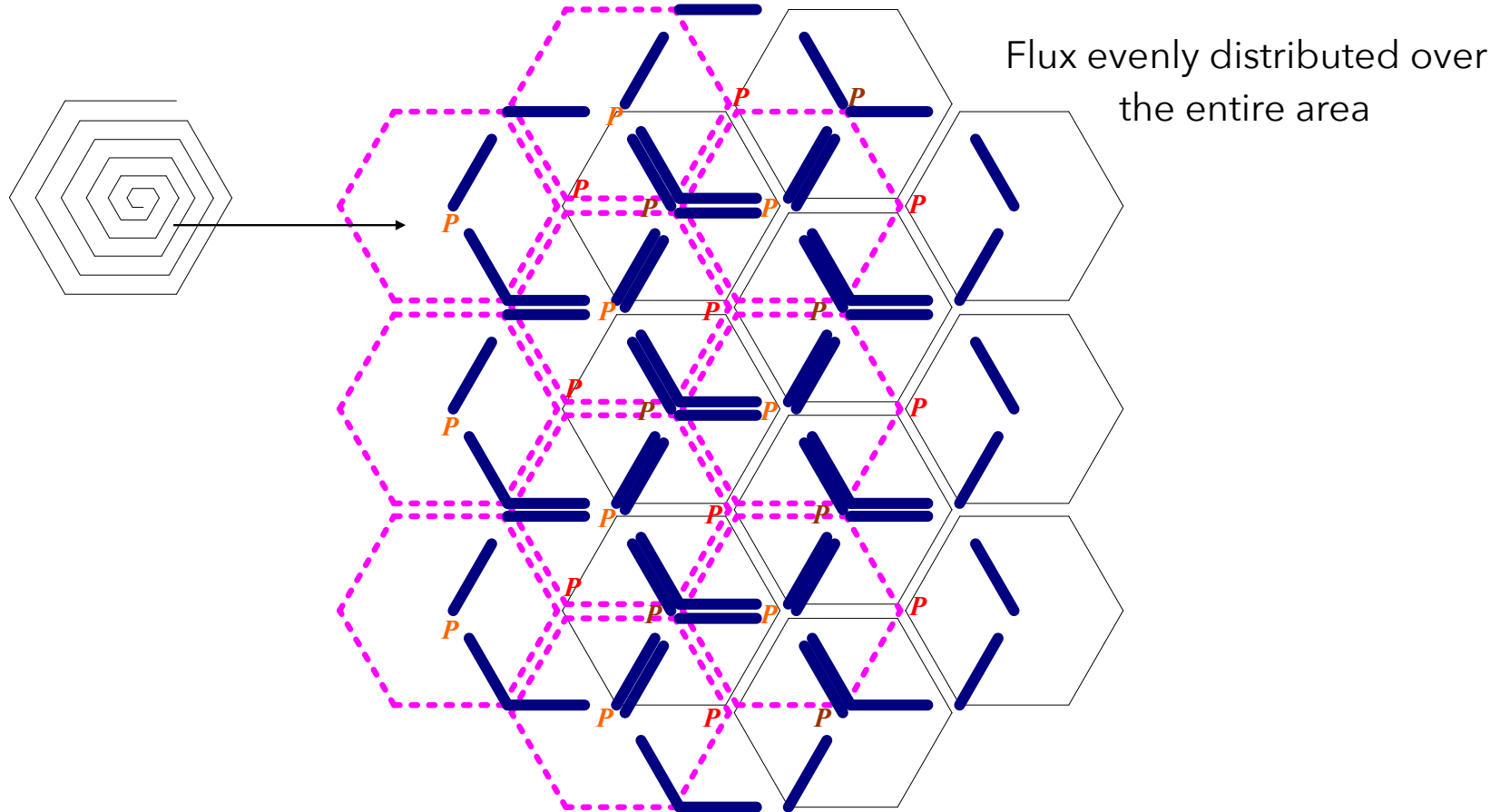
5 kW



30 kW

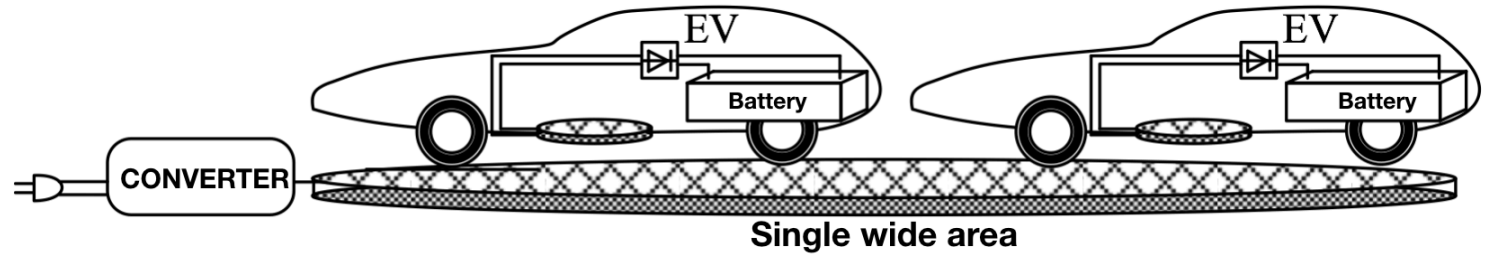


3-layer PCB windings

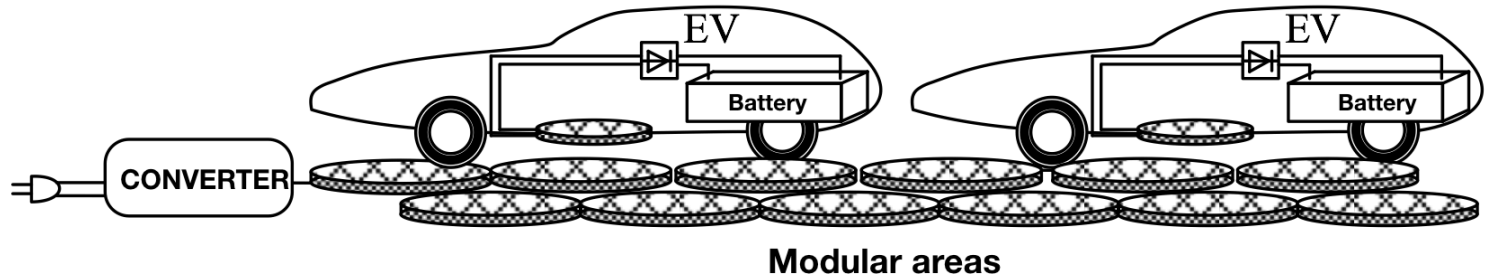


Sectional charging pads

Wide area coverage

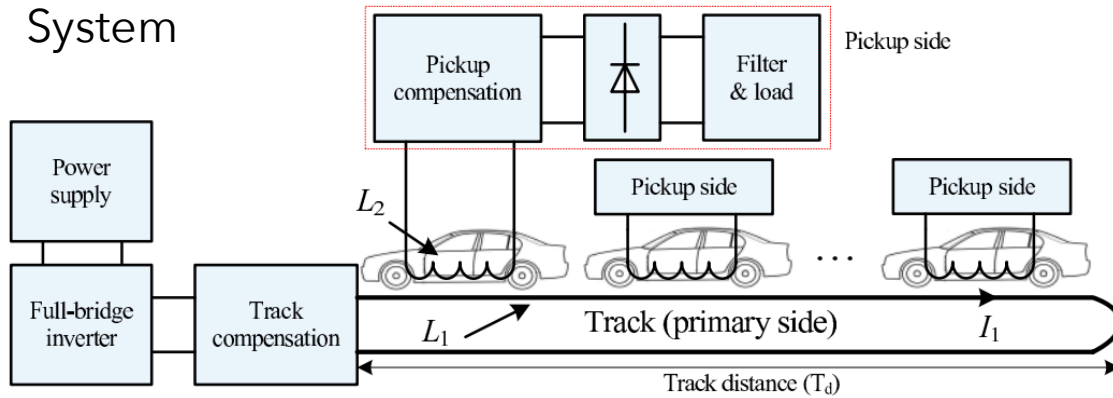


Cost and material reduction,
still covering wide area

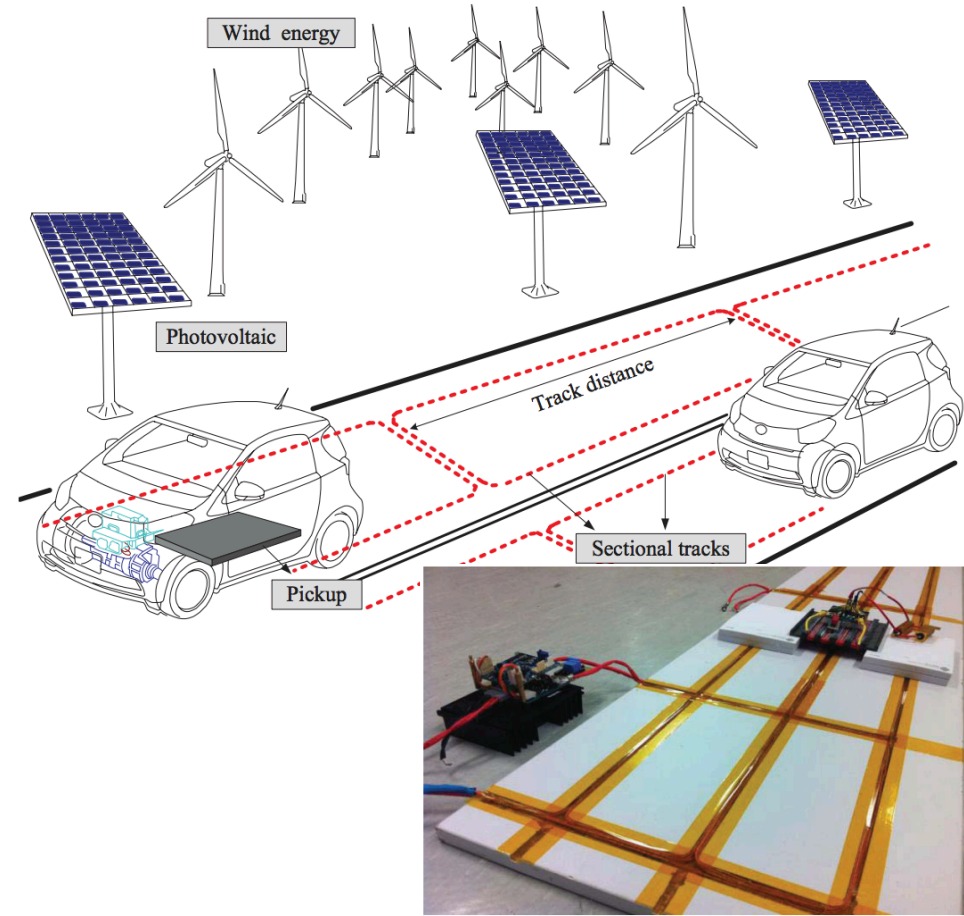
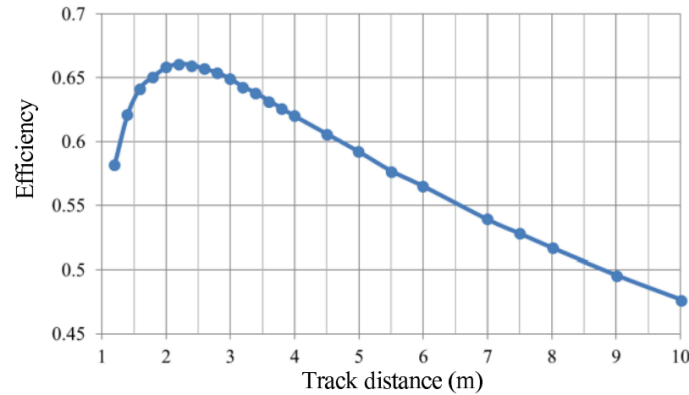


Roadway charging

Sectional Tracks in Roadway Inductive Power Transfer System



Analysis for moving vehicles charging system, multiple vehicles sharing power, distance, etc.



Interim Conclusion

- So far, the circular pads are still the most popular.
- DD, DDQ pads are receiving significant attention as they can increase coupling capability.
- Lots of results have been obtained mainly from EXPERIMENTAL studies.
 - Coupling
 - Misalignment
 - Efficiency
 - Cost
- Analysis is possible, but not very comprehensive due to the complex geometry and flux interaction. A conventional EM field.
- New core geometries and winding arrangements will continue to emerge in the coming years. Lots of opportunities for research.